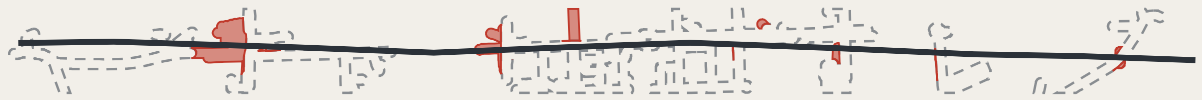
A civic AI leveling network that uses closure error as its trust metric

Leveling is not about measuring accurately; it is about measuring back.
Depart, carry the height station by station, return — that gap is f.
Within tolerance, every station on the line is accepted;
over tolerance the whole run is void; no station may be patched alone.

That hundred-year-old rule answers this belt's hardest question:
what a city's trust in its AI rests on: not performance, but return.



Ground within a ten-minute walk of a benchmark

Dashed: 232 buildings today. Filled: the ground that only comes within ten minutes once the eleven stitching links and three spurs are built — 261 in all, a gain of 29.
Catchment 203.3 ha to 220.6 ha; 800 m of network distance, not a radius.

READINGS, ALL RECOMPUTED FROM THE SHIPPED FILES

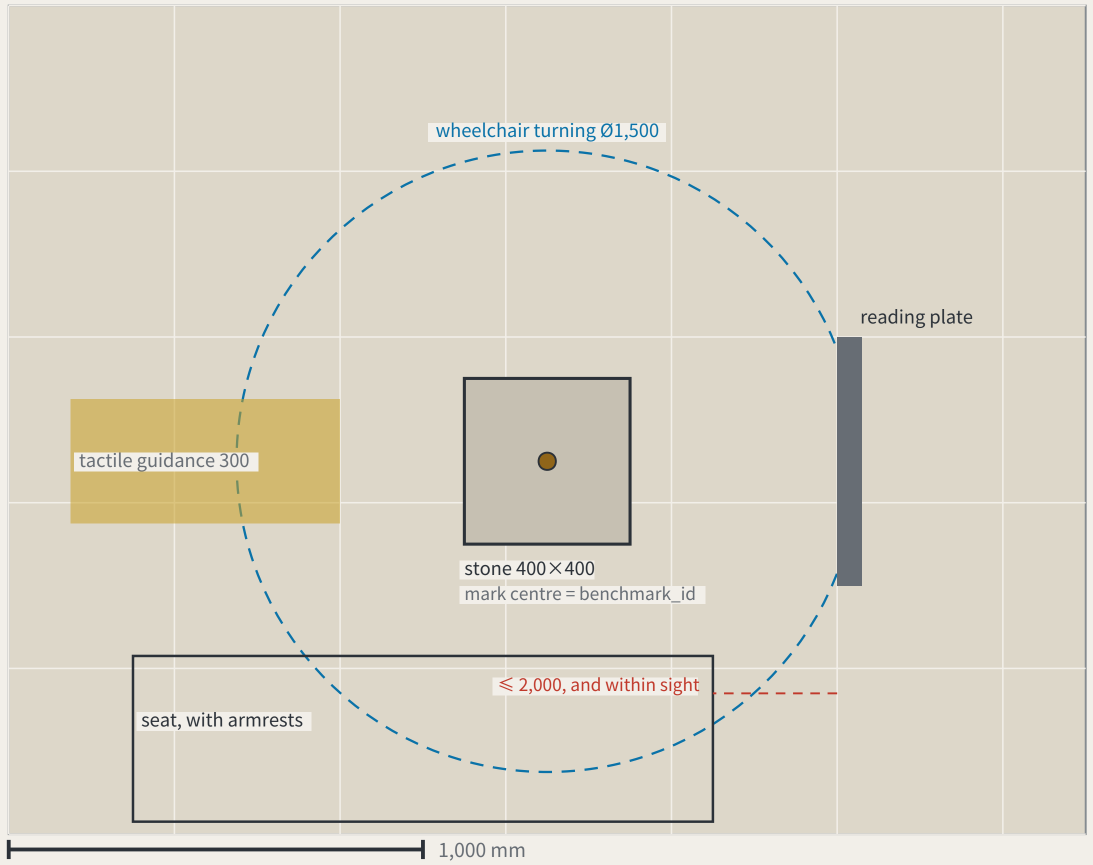
11,412,825 m²	Overall design area
9,443 m	Spine length
8	Tiered benchmarks
13.8 km/km²	Network density, measured
17	Oversized blocks, not parks
19.7%	Green corridor, share of design area

What this proposal does not give here matters as much as what it does

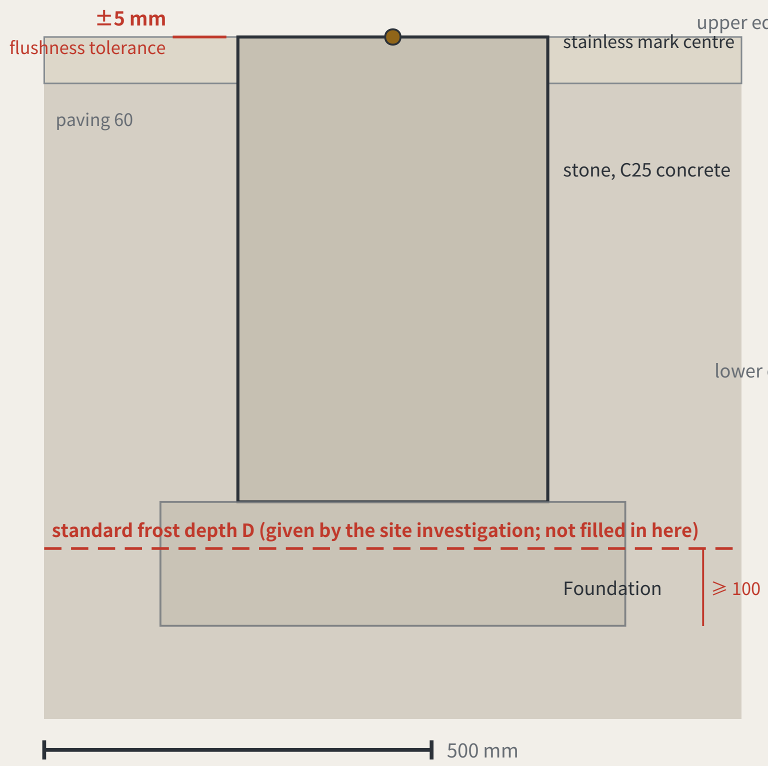
Plot ratio, height, coverage, setbacks, road redlines — inside the statutory approval process; none is drafted here before the official conditions are published.
The tolerance values per scenario — set by the tolerance assembly at BM-1; this proposal gives the grading basis and drafts no value.
Any engineering feasibility conclusion — stitching points register a need for connection; transport and structural review is outside this proposal.

a type drawing; it carries no orientation

I. Plan

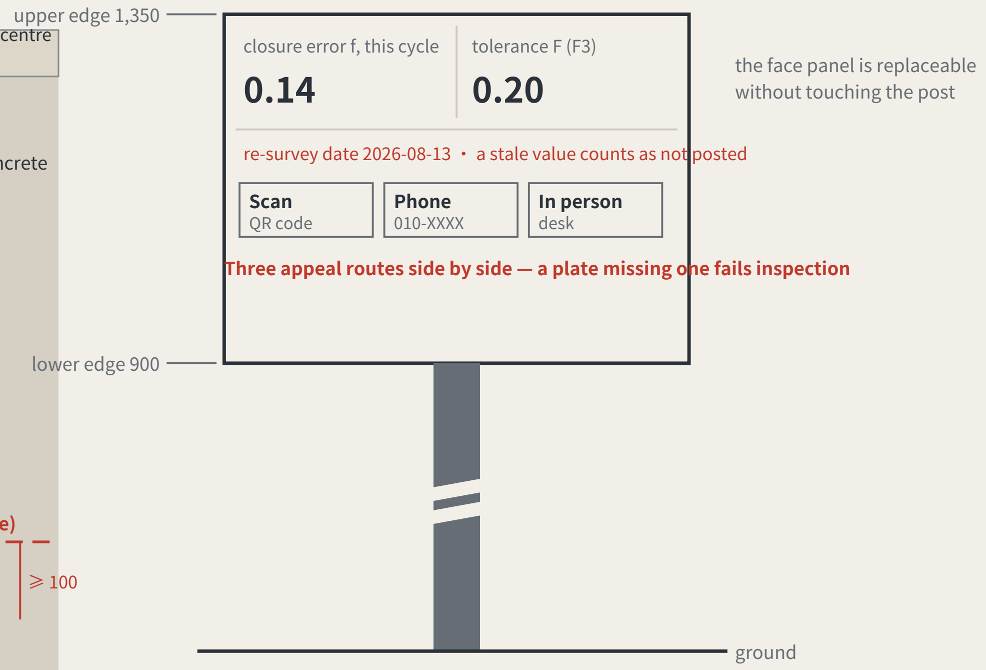


II. Section



III. Plate elevation (post shown broken)

Specimen values; F is a dimensionless rate, read from scenario_cards.json



Dimensions: design intent and as built

The right column is deliberately empty — as-built values are measured on site

Item	Design intent	Why this number	As measured
The stone	400 × 400 × 600 mm, C25 concrete, stainless mark centre in the top face	the mark centre is the point a benchmark_id refers to	
Stone top against the paving	⩽ ±5 mm	KIT-01 says flush with the ground and no trip hazard; without a tolerance, “flush” is an opinion	
Depth to underside of foundation	⩾ the local standard frost depth D + 100 mm	D comes from the site investigation and is not filled in here — this package holds no verifiable frost-depth source for Haidian	
Plate face size	600 × 450 mm, raked 15°	the rake cuts glare and standing water; the face is replaceable without the post	
Plate lower / upper edge	900 mm / 1,350 mm	the band a seated and a standing reader share; P5’s continuous step-free route should not end at a plate they cannot read	
Appeal panel	QR code, phone number and in-person address, all three side by side	KIT-05: a QR code alone excludes anyone without a smartphone from appealing, which makes persona P4 unworkable	
Tactile guidance strip	300 mm wide, stopping 300 mm short of the stone	the stop line leaves room to stand and take a reading rather than leading onto the stone	
Wheelchair turning space	Ø1,500 mm, headroom ⩾ 2,100 mm	the turning space coincides with the reading position, so taking a reading needs no reversing first	
Seat	seat 450 mm high, with armrests, within 2,000 mm of the plate	KIT-03: taking a reading should not require standing and waiting, and the seat must be in sight of the plate	

The whole proposal rests on one object, and not one dimension had appeared anywhere in the package — a component library without dimensions is a mood board. Here it is in plan, section and elevation, nine dimensions each with the reason it is that number; read the scale off the bars. The one figure left blank is the frost depth D: no verifiable source here, so the sheet gives the rule and not the number. The as-built column is empty too. Concept design intent; requires detailed design and engineering acceptance.

a type drawing; it carries no orientation

I. Order of removal: from what is swapped to what is never moved

One person is assumed to carry up to 23 kg. The stone, 400 × 400 × 600 mm of plain concrete at 2,400 kg/m³, works out at about 230 kg — the one part nobody can carry is exactly the one part that should never be replaced.

II. What fails, how often it is replaced, and whether it touches the datum

Part	How it fails	Replaced	Mass	Tool	Touches datum
Face panel of the reading plate	stale values, scratching, fading	checked each cycle, replaced when damaged	3.0 kg	hex key	no
QR and appeal sticker	the link dies, the surface wears	annual	0.2 kg	none	no
post	impact, corrosion	about 20 years, or after an impact	12.0 kg	socket wrench	no
seat panel	wear, graffiti	about 10 years	18.0 kg	socket wrench	no
Tactile guidance strip	abrasion, and repaving	about 15 years, with the paving	6.0 kg	paving tools	no
The stone	impact, excavation, settlement	**never replaced**	230 kg	plant required	**yes**

III. After it fails: found → held → replaced → lost

Found any re-survey that finds the face stale or missing records it in that cycle
the plate's own rule — a stale value counts as not posted — is the detector

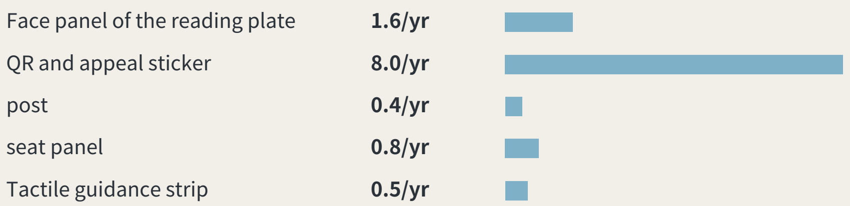
Held the point is marked returned in datum red, shown at every third-order point in the area
bad news belongs where it happened (errata E50)

Replaced a standard part is swapped without touching the stone or its mark centre
one person, one visit; heaviest part 18 kg

Stone lost the height is re-established from the next order up; the closure record carries the break; the id is retired and never reused
reusing an id would quietly join two different points into one history

IV. What these intervals add up to

Summing the intervals above across 8 benchmarks: about 1.42 replacements per point per year, roughly 11 across the line, about 6 staff-hours a year. Those hours were not in the annual operating table — it counted re-survey sessions only. They are now: paid hours go from 133–214 h to 139–220 h, adding about CNY 680–1,473 a year. Maintenance is not a hidden cost, but it genuinely was not in the table.



Intervals and hours per swap are assumptions, not measurements; the stone is never replaced and is excluded.

A monthly re-survey is a maintenance problem first, and this package had said nothing about what fails, how often, or who carries the replacement. One asymmetry organises the sheet: the stone is the only part that must never be replaced and the only one a person cannot carry (about 230 kg) — a benchmark that has moved is not the same benchmark. So the wear goes on the swappable parts. Intervals and masses are estimates, not measured.

Method: trust comes from closure, not from one accurate reading

Leveling is not about measuring accurately; it is about measuring back. Depart from a known point, run the circuit, return — the difference is the closure error. Over tolerance the whole run is void and re-measured, and you may not patch the worst station and keep the rest.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and autonomous shuttles, and AI health, education, legal and daily services. Urban AI governance is the method layer here, not the selling point.

Two prohibitions at the heart of the mechanism

One: amending only the worst station is not allowed. This makes tuning until the metric looks good structurally ineffective.

Two: tolerance may tighten on evidence and may never loosen because a scenario failed. This closes off the moving goalpost.

f is always defined as a deviation, never an attainment score: classification scenarios take 1 minus the consistency ratio, service scenarios the satisfaction range, safety scenarios the sum of false-positive and false-negative rates. All three run the same direction, so $f \leqslant F$ is always the passing test.

Act One: measure this open call first

228 Merged proposals (git tree, authoritative)

30.3% Machine-readable disclosure field empty

84 → 8 Distinct strings written → model families

The "machine-readable" disclosure field cannot actually be aggregated: the GPT/Codex family alone is written 44 ways across 103 proposals. The fix is light — split the field into an enumerated family plus free-text detail, and add one check to the gates.

Closure error in the site's own data

412.5 m Provisional design area to surveyed park

100% Agreement with the research area

Checking the inferred boundary against OpenStreetMap's surveyed heritage park, two independent routes to the same object do not close. The 43.6 km² level agrees completely; the discrepancy is confined to the 11.4 km² level.

The same measurement also measured this proposal: the submitted spine lies 1,116.7 m from the surveyed park, because it was generated against the provisional boundary. This is stated because a recomputability standard invoked only when it flatters the author is not a standard.

Main front one: low-speed robots and autonomous shuttles

No benchmark, no robot. This turns governance into a spatial design problem: to expand the operating area you must first build measurement points.

Across the field, speed limits, remote and physical emergency stop, on-site safety officers, incident logs, an equivalent non-AI path and scenario-level exit are already the de facto standard. This proposal adopts all of them but presents none as an innovation — they are the admission floor.

The increment is where nobody has been: ice and low-temperature re-survey (zero hits across the field), noise as a number (zero hits), jurisdictional seams, fleet density ceilings, and emergency-access yielding. Their shared structure is that the same system behaves inconsistently across conditions or jurisdictions — which is exactly what closure error measures.

Any safety incident suspends that entire machine type network-wide, not the individual unit. Tolerance scales with kinetic energy: faster or heavier requires a stricter closure clearance first, never an exemption.

Jurisdictional seams: where pilots actually die

8 / 8 Cross-jurisdiction points on this belt

Crossing jurisdictions is the normal condition, not an edge case. The failure mode at a seam is not a contest over authority but that each side reasonably concludes it is not theirs, so the device keeps running unreviewed until something happens. The rule: if neither side holds a valid reading, the section is closed — so the default consequence of inaction is that the device stops.

Main front two: AI public services

Do not measure the average; measure the dispersion. The risk is not in mean accuracy but in how much the conclusion differs when different people ask the same question — which is precisely what closure error measures.

Three non-waivable boundaries: prescriptive judgements must be made by qualified people and logged; the equivalent non-AI path is permanent and must not be slower; and no profiles of identifiable individuals and no cross-scenario linkage.

Statement of boundaries

Everything here is open collaborative concept advice on provisional substitute boundaries. It does not replace statutory planning and constitutes no government determination, approval basis or implementation commitment. No floor area ratio, building height, density, setback or road redline figures are given, and no heritage protection zone or blue line is inferred. Responsibility is written as role types only, all subject to negotiation. Final judgement rests with people and professional teams.

THE LEVELING LINE

THE LEVELING LINE

A city that publishes its own error.

A city that publishes its own error.

Leveling is not about measuring accurately. It is about measuring back.

Depart from the origin, carry the height station by station, run the circuit and return — the part that does not land back on the datum is the closure error.

Within tolerance every station is accepted; over tolerance the whole run is void and re-measured, and no single station may be patched.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and AI public services.

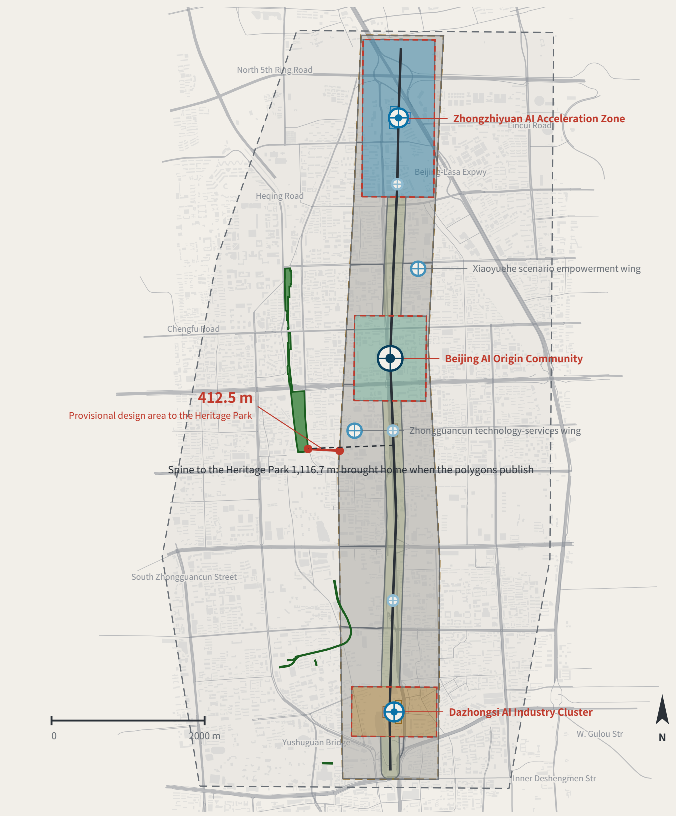
OVERALL CONCEPT AND SITE OVERVIEW

FIG.01

Research area agrees 100% with the surveyed park; the gap is in the design area only, nearest 412.5 m — and this proposal's spine is displaced 1,116.7 m with it.

north is up

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



Legend and how to read it

LEGEND • surveyed data and inferred boundaries are shown separately

Surveyed (independently checkable)

- Jing-Zhang Railway Heritage Park (OSM survey, 17.49 ha)
- Disused railway (14 segments, 3,028 m)
- Existing street network (OpenStreetMap, 1,376 ways in this window, expressway to tertiary; local lanes not drawn)
- Existing building footprints (OpenStreetMap, 7,709 in this window)

Inferred (not usable as a redline)

- Coordinated research area 43.6 km²
- Overall design area 11.4 km²

This proposal's design (generated against inferred boundaries)

Blank-block edges follow the existing arterials; they are not a parcel subdivision or an ownership boundary

- Cultural use (0803)
- AI R&D and research (0802)
- Community services and support (0702)
- Industry and commercial services (05)
- Park / green (1401)
- Urban road land (1207)
- Left blank by this proposal (code 16)
- Spine (design centreline)
- Tiered benchmarks

The line to read on this sheet is the red one.

Research area and surveyed park agree 100%, so the announced bounds do not conflict with actual geography; the discrepancy is confined to the overall design area.

This proposal's spine is displaced with it. Marked here rather than hidden.

What the belt does

WHAT THE BELT DOES • the five functions are five positions on one loop

Set the tolerance	Depart	Take the readings	Close at the origin	Over tolerance: re-survey
BM-1 Zhongzhiyuan	BM-0 Origin Community	BM-2 Dazhongsi, the wings	BM-0	the whole route returns

Over tolerance the whole route returns for re-survey and drops to the non-AI equivalent - the loop does not close, and the belt does not run.

9.4 km — Spine length

11.4 km² — Overall design area

8 — Tiered benchmarks (four orders)

Site cross-check (the red line at left):

The announcement draws the design area 1-2 km around the Jing-Zhang Rail Heritage Park - public ground already open. That park, from OSM, is 17.49 ha: 100% of the research area, 0% of the design area, nearest 412.5 m.

Same instrument on this proposal: the spine sits 1,116.7 m from that park - the distance to be brought home when the official polygons publish.

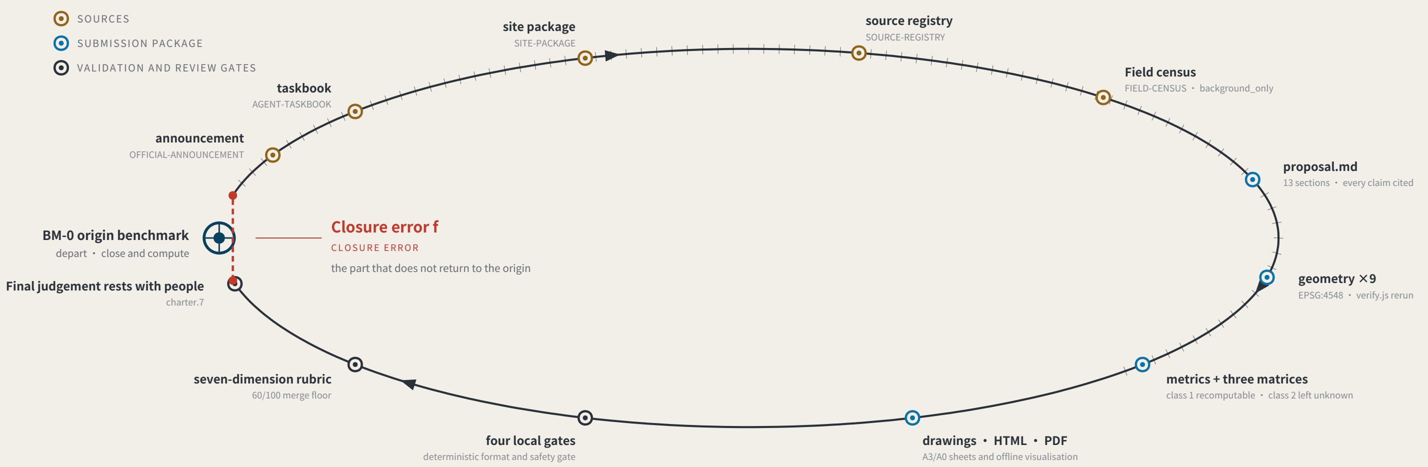
A recomputability standard invoked only when it flatters the author is not a standard.

Surveyed data © OpenStreetMap contributors • ODbL 1.0 • Overpass API 2026-08-08. Script and conventions in visual/assets/osm_reference.json; re-runnable. OSM is crowd-sourced and the park polygon may cover only the mapped section; the provisional boundary is itself inferred. This sheet reports the difference and does not adjudicate which is right.

EVIDENCE CHAIN AS AN UNCLOSED LEVELING CIRCUIT

FIG.02

a type drawing; it carries no orientation



Act One: measuring this open call itself

MEASURING THE OPEN CALL ITSELF • as of 2026-08-13 • 793 merged • 793/793 fetched • PR numbers past 1000

793
Merged proposals (git tree, authoritative)
The gallery index lists 507 at the same moment — 287 behind

18.2%
Machine-readable disclosure field empty
144 of 793 still hold the scaffold placeholder or are empty

228 → 9
Distinct strings written → actual model families
The GPT/Codex family alone is written 85 ways across 355 proposals

The "machine-readable" disclosure field is not machine-readable.

charter.6 requires disclosing the generation method and charter.5 requires it structured and agent-readable, yet none of the four gates checks this field. Nobody can aggregate "which models produced this call" from it.

The fix is light: split `model` into an enumerated family plus free-text detail, and add one enumeration check.

This proposal is the instrument for the missing last step: compute the closure error, and give it consequences.

Registers answer "what did the system declare" and assessments answer "what do experts think"; closure error answers "how much does the conclusion differ across nodes and across time for the same public question".

Sources: visual/assets/census.json, field_map.json and the first-round (184-file) snapshot agent_declarations.json, all shipped with the package. The authoritative source is the GitHub git tree, not the gallery index, which lags. Generation scripts are in the accompanying issue and are re-runnable. The corpus grows daily; recompute before citing.

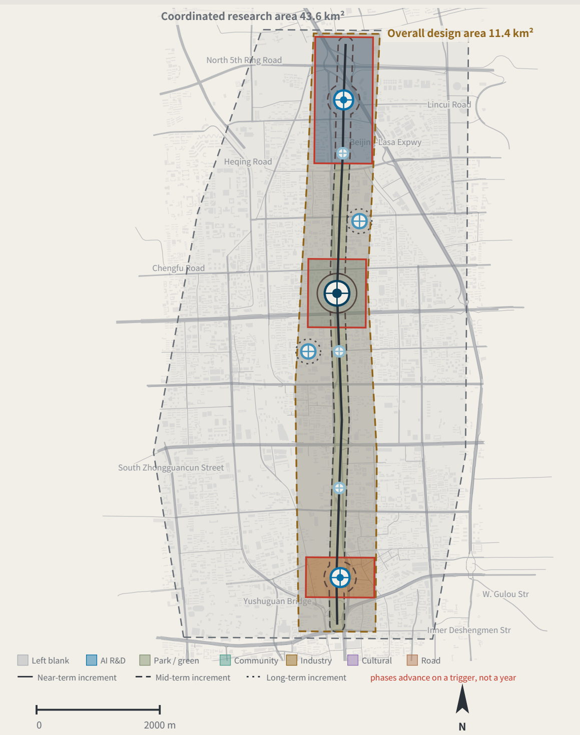
LAND-USE STRUCTURE AND THE THREE SCOPE LEVELS

FIG.03

7 land-use classes, no overlap and no gap, verified point by point; the 55.4% left blank is a judgement — tenure and control plans are unpublished.

north is up

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



Plan, north up. All boundaries are provisional substitutes; official polygons are unpublished.

Any area, ratio or count that cannot be recomputed from a structured layer **is not written as a conclusion.**

THREE SCOPE LEVELS MAPPED ONTO SURVEY ORDERS

SCOPE LEVELS AND NETWORK TIERS

Announcement level	Scale	Network role	Re-survey cycle
Coordinated research area	43.6 km²	Whole-network control	**Annual**
Overall design area	11.4 km²	First-order route: spine plus two closing routes	**Semi-annual**
Key areas	368.4 ha	Origin BM-0 with first-order BM-1 / BM-2	**Quarterly**

The three levels are not three unrelated drawing sets

The research area decides what to measure.

The design area decides which route to measure along.

The key areas decide where to set the stones.

Benchmark orders

BENCHMARK ORDERS

Origin benchmark	Annual origin and closure check
First-order benchmark	Annual re-survey
2nd-order point	Quarterly re-survey
3rd-order point	Monthly re-survey

Land-use partition composition

LAND-USE PARTITION 22 features • 7 classes • no overlap, no gap (verified point by point)

Reserved (16)	55.4% • 11 blocks
AI R&D (0802)	16.9% • 2 blocks
Park / green (1401)	12.2% • 1 block
Community services (0702)	8.6% • 1 block
Industry and commerce (05)	6.3% • 2 blocks
Cultural (0803)	0.6% • 4 blocks
Road (1207)	0.1% • 1 block

Reserved land at 55.4% is a judgement, not a gap: tenure, plan and retain/demolish status are unpublished; this proposal does not subdivide for them.

The seven shares are each rounded to one decimal and the column sums to 100.1%; the partition itself is verified point by point, with no overlap and no gap.

Values this proposal deliberately does not give

Floor area ratio • building height • density • setbacks • road redlines

Statutory regulatory-plan controls. Publishing a figure with the official data absent would manufacture certainty; metrics.json keeps them unknown.

Layers: brief/site-package/geometry/provisional_boundaries.geojson, geometry/roads.geojson, public_space.geojson, land_use.geojson, phasing.geojson. Drawn by the accompanying scripts and recomputable (scripts in the accompanying issue). The base street network and building footprints are measured OpenStreetMap data (ODbL 1.0), drawn for location only and not part of this proposal's design. The three areas are taken from the announcement text; the geometry is a provisional substitute.

THE THREE KEY AREAS, THE TWO WINGS AND THE BENCHMARK LAYOUT

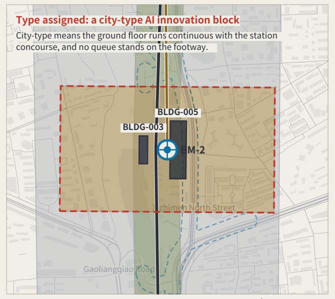
FIG.04

orientation is noted on each drawing

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

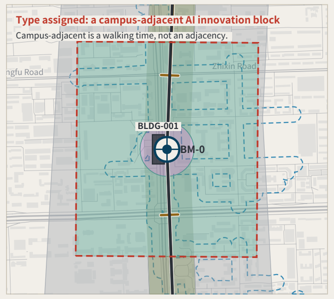
Dazhongsi AI Industry Cluster

BM-2 • first order



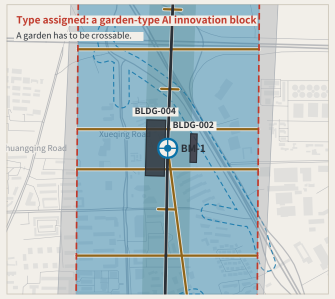
Beijing AI Origin Community

BM-0 origin • network start and closure • evidence hall, stone L1



Zhongzhiyuan AI Acceleration Zone

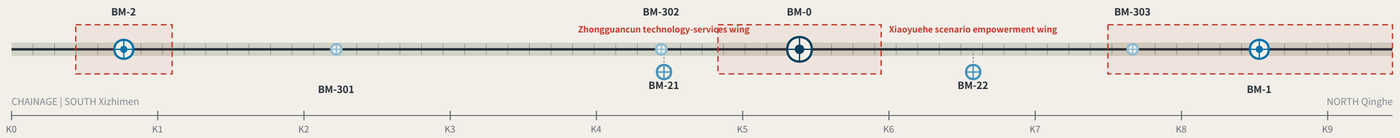
BM-1 • first order



0 — 500 m; N

Land-use class: Reserved (16) · AI R&D and research (0802) · Community services and support (0702) · Industry and commercial services (05) · Cultural use (0803) · Park / green (1401)

Brass: routes this proposal draws; grey beneath: surveyed OSM street. The olive band is the continuous corridor layer, 224.41 ha — **not** class 1401, whose 139.23 ha is that corridor folded into the partition. Three windows, one scale, north up. Each frame is 1,500 m N-S by 1,667 m E-W; the tighter axis sets the scale, so 167 m more ground shows across. Zhongzhiyuan runs 2,061 m N-S past its frame — see the locator. Dashed lines provisional.



Benchmark orders

BENCHMARK ORDERS

- **Origin benchmark** — Annual origin and closure check
- **First-order benchmark** — Annual re-survey
- **2nd-order point** — Quarterly re-survey
- **3rd-order point** — Monthly re-survey

Closure verdict

CLOSURE RULE

A scenario departs BM-0, is read at each station along a connecting route by a different party, and terminates at a first-order point; the computation closes at BM-0.

- f ⩽ F level for this cycle: it may operate and enter the next cycle
- **f > F the whole route returns for re-survey and drops to the non-AI equivalent**

Amending only the worst station is not allowed. Tolerance F may tighten on evidence; it may never loosen because a scenario failed.

Three areas, two wings and survey roles

THREE AREAS, TWO WINGS AND SURVEY ROLES

Dazhongsi AI Industry Cluster High-frequency reading point | AI-native consumption and business	BM-2 • first order
Beijing AI Origin Community Network origin and closure computation | public evidence hall and stone L1	BM-0 origin benchmark
Zhongzhiyuan AI Acceleration Zone Tolerance datum | where F is set	BM-1 • first order
Zhongguancun technology-services wing Support system | capital and factors; takes no reading of its own	BM-21 second-order benchmark
Xiaoyuehe scenario empowerment wing Where the stations are densest | real users	BM-22 second-order benchmark

A wing carries a second-order point, not an area design; its requirement is written as dimensions, on FIG.26.

Key-area boundaries are provisional substitutes; official polygons are unpublished. Dashed; not a redline or an area basis.

Layers: geometry/key_areas.geojson, roads.geojson, public_space.geojson, green_space.geojson. Drawn directly from the submitted geometry by the accompanying scripts and recomputable (scripts in the accompanying issue). In all three windows the base street network and building footprints are measured OpenStreetMap data (ODbL 1.0, the same cache as the density readings), drawn for location only and not part of this proposal's design. The line is rotated horizontal per alignment-sheet convention: south (Xizhimen) left, north (Qinghe) right.

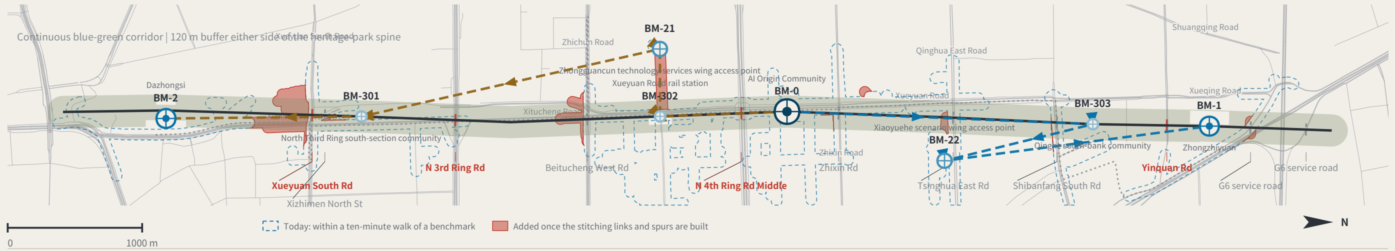
SLOW MOBILITY, BLUE-GREEN AND CONNECTING ROUTES

FIG.05

Six spine benchmarks: 11,472 m walked today against 9,443 m drawn (x1.21); with the stitching and spurs built, buildings within a ten-minute walk rise 232 to 261.

north is right (unrolled)

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



East-west stitching points (11)

EAST-WEST STITCHING POINTS

The class rests on one measurable fact: whether OSM already maps a crossing within 60 m of the intersection. If it does, Class A. If it does not, the need is registered and left ungraded — separating Class B (channelisation) from Class C (new structure) is a traffic study's conclusion, not this proposal's.

Yinquan Rd	Class A	10 m	**Xueyuan South Rd**	Class A	25 m	**N 4th Ring Rd Middle**	Class A	30 m	**N 3rd Ring Rd**	Class A	33 m
Beitucheng West Rd	need registered only	65 m	**Xizhimen North St**	need registered only	68 m	**Tsinghua East Rd**	need registered only	86 m	**Shibanfang South Rd**	need registered only	146 m
G6 service road	need registered only	166 m	**G6 service road**	need registered only	175 m	**Zhixin Rd**	need registered only	201 m			

Distances are to the nearest mapped crossing. OSM is crowd-sourced: "need registered only" means OSM shows none, not that none exists on the ground.

The two connecting routes

CONNECTING ROUTES

North connecting route RT-N
BM-0 → BM-303 → BM-22 → BM-1 (terminates at a first-order point; the computation closes at BM-0)

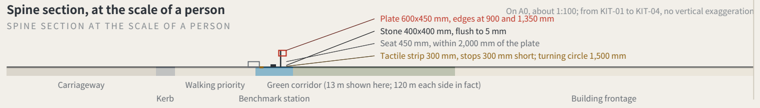
South connecting route RT-S
BM-0 → BM-302 → BM-21 → BM-301 → BM-2 (terminates at a first-order point; the computation closes at BM-0)

Walking the six spine benchmarks on the mapped network, measured
11,472 m against 9,443 m: 2,029 m further today, ×1.21 — that difference is what the belt is for
Two of the eight are outside this chain. BM-21 has no walkable path to any other benchmark on this graph at all — not a detour, unreachable. BM-22 is reached by a spur and is registered separately, so that one leg is not counted twice.

Each station is read independently by a different party — professional body, operator, resident representative, international visitor, all four required. Homogeneous review parties would bias the closure error systematically small and rob it of meaning, a risk this mechanism declares.

Spine section, at the scale of a person

SPINE SECTION AT THE SCALE OF A PERSON



SLOW MOBILITY AND THE BLUE-GREEN NETWORK

SLOW MOBILITY AND BLUE-GREEN

Spine	A continuous walkable public axis; walking and cycling take priority	9,443 m
Green corridor	Buffered both sides of the spine, forming a continuous blue-green interface	120 m each side
Station = benchmark	Rail concourses double as third-order benchmarks; re-survey where footfall is densest	3
Universal first	Step-free access, lighting, drainage and shade come before any smart facility	Precondition

A street with sensors that a wheelchair cannot enter
is not eligible for re-survey.

Tolerance classes

TOLERANCE CLASSES

F1	**strictest**	Bodily safety or administrative acts	S04 • S06 • S10 • S11
F2	**Medium**	Individual rights	S02 • S03 • S05 • S09 • S12
F3	**loosest**	Information only	S01 • S07 • S08

Layers: geometry/roads.geojson, green_space.geojson, public_space.geojson. Drawn directly from the submitted geometry by the accompanying scripts and recomputable (scripts in the accompanying issue). Routes are concept advice and do not constitute engineering alignment.

RECOMPUTED METRICS AND CLOSURE EVIDENCE

FIG.06

Run once: 10 cases, 2 accepted, 8 refused, reasons quoted from shipped files; 12 cards name the path that remains with the AI off.

a type drawing; it carries no orientation

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

METRICS RECOMPUTATION

RECOMPUTED METRICS • EPSG:4548

Class 1, measured: recomputable directly from this package's geometry

Overall design area	11,412,825 m²	Phase increments, summed	2,920,311 m²
Spine length	9,443 m	Phased share of design scope	0.2559
Benchmark count	8	Existing street length, measured	157,510 m
Green corridor ratio	0.1966	Existing network density, m/m²	0.0138
Public measurement-point ratio	0.0642	Existing median block, measured	4,300 m²
Indicative footprint area	82,276 m²	Existing building coverage, lower bound	0.1798
Key areas	3	Six spine benchmarks walked, mapped net	11,472 m
Green ratio (land-use partition basis)	0.1220		

A further 31 count metrics (unit: count) ship in metrics.json:
cards, personas, landmarks, cases, projects, phases, land-use classes, errata — all recomputed by the build, none in this table.

Class 2: requires official regulatory support; held at unknown

Floor area ratio • building height • density • setbacks • road redlines

Filling estimates into a gap while official data is absent is fabricated certainty.

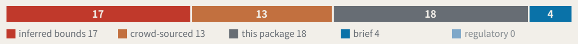
Class 3: requires continuous re-survey calibration; no baseline yet

Per-scenario closure error f • tolerance compliance rate • non-AI path coverage • resident-initiated re-surveys

Baselines follow one near-term cycle. This proposal says no data exists rather than passing intent off as measurement.

Evidence ceiling: computed exactly ≠ measured reliably

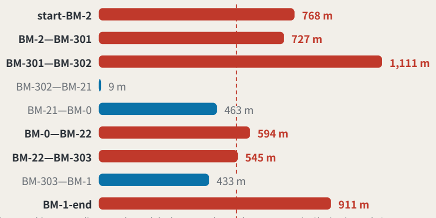
Each metric's ceiling is the weakest of the files it reads, computed by the build rather than judged by the author



53 metrics, 52 of them gradeable; 17 are drawn on an inferred boundary and 0 reach the regulatory baseline.
Every class-1 value is recomputed from the shipped layers by the accompanying script;
A number nobody can recompute is not evidence — and that standard applies to this proposal first.

Walk to the nearest benchmark: 9 segments, 6 of them failing

WALK TO THE NEAREST BENCHMARK • worst per segment; red line 540 m (P5 at 0.6 m/s, 15 min)



Worst 1,111 m; for P5 that is 30.9 minutes - this proposal's own rule, and the layout as drawn does not meet it. Closing it needs 9 more third-order points, whose annual operating cost is already in this package's cost table rather than left to a next version.

Closure evidence and measured corpus

CLOSURE EVIDENCE AND MEASURED CORPUS • n=793 (git tree) • 2026-08-13

Closure evidence: the mechanism has been run

`node visual/assets/run_s08_tabletop.js` • 10 cases • 2 accepted • 8 refused • re-runnable by the reviewer • rule and reason quoted verbatim from the shipped file

- **accept** f <= F is level for this cycle and the scenario continues
- **accept** f > F returns the whole route, not the failing station
- **refuse** rule 5: amending only the worst station is not an available action
  record.stations: needs at least 3 items, has 2; review parties: 2 distinct across the circuit; a reading taken by one kind of party cannot detect what that kind of party cannot see
- **refuse** a record does not get to state its own f
  verdict.f declares 0.13 but the stations recompute to 0.360000; verdict.passed says true but f 0.360 against F 0.2 derives false; verdict.action is "continue" but f against F requires "return-whole-route"
- **refuse** a circuit read by one kind of party cannot detect what that party cannot see
  review parties: 1 distinct across the circuit; a reading taken by one kind of party cannot detect what that kind of party cannot see
- **refuse** without an equivalent non-AI path the stop rule would be circumvented
  record.verdict.non_ai_path_available: must be true, got false; verdict.non_ai_path_available is false: a scenario with no non-AI equivalent cannot be returned, because returning it would interrupt public service — so the stop rule would be circumvented
- **refuse** omitting the non-AI path field is not the same as satisfying it
  record.verdict: missing required property "non_ai_path_available"; verdict.non_ai_path_available is absent: the equivalent non-AI path is not waivable, so omitting the field cannot be treated as satisfying it. State it explicitly as true.
- **refuse** a connecting route must terminate at a first-order benchmark, or nothing checks the carried value
  stations[last]: a connecting route must terminate at a first-order benchmark, because the carried value has nothing to be checked against otherwise. This one ends at BM-302 (third).
- **refuse** rule 7: two consecutive passes are required, because once may be luck
  resumption needs two consecutive passing cycles, record shows 1; once may be luck
- **refuse** rule 7: an F1 incident resumes only on unanimous confirmation, no majority rule
  resumption after an F1 incident requires unanimous confirmation by all four review categories

Nothing here is a field measurement. Not one reading was taken: these are worked values chosen to exercise the refusal branches.

Measured corpus: few take this road

793 packages, read from the git tree (507 listed in the index). Pulse or lifeline is at 168 and proof or verification at 184, both saturated; measurement or levelling is at 10. Motif detection is a keyword regex and therefore a lower bound, and the convergence on three-areas-two-wings is induced by the taskbook rather than agreed.

Leveling is not another railway-operations metaphor. It is a method for how trust is established.

What is left of the belt with the AI switched off

Without a non-AI equivalent the stop rule is circumvented. All 12 cards name their equivalent; the 4 F1 ones keep theirs permanently (full text in scenario_cards.json).

S01	F3	Paper sign-in and on-site stewards	S02	F2	Residents report breaks on paper at the nearest third-order benchmark,...	S03	F2	The staffed counter runs independently of the agent
S04	F1	A staffed help desk and a printed directory of institutions, departments and clinic hours	S05	F2	Authorisation credentials and change records are documents	S06	F1	Devices stop and return to their bays
S07	F3	The evidence hall's physical display and paper archive stay open	S08	F3	Physical interpretation plates along the route and scheduled human-led tours	S09	F2	Prices and licences go on a physical notice board
S10	F1	A review meeting and a paper record	S11	F1	The test ground stops serving the public and its boundary is held by physical...	S12	F2	Checking reverts to user reports on paper or by phone plus quarterly...

Sources: this package's metrics.json and visual/assets/field_map.json, both shipped with it; the authoritative source is the GitHub git tree. The independent verifier visual/assets/verify.js checks every class-1 metric. Generation scripts are in the accompanying issue. The corpus grows daily; re-run before citing.

IDENTITY: MARK, CONSTRUCTION AND APPLICATIONS

FIG.07

a type drawing; it carries no orientation

THE LEVELING LINE

THE LEVELING LINE

The mark draws the method, not decoration: the datum line departs the origin, rises, returns, and misses the datum - that red gap is the closure error.

Construction

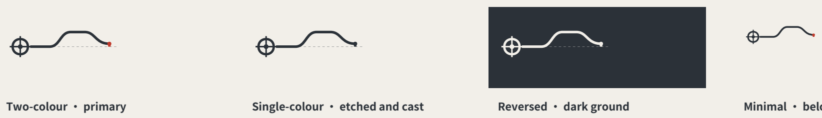
CONSTRUCTION

- **Origin benchmark** — Ring and crosshair; the network's starting point
- **Datum line** — Dashed is the ideal datum; solid is the measured route
- **Rise and return** — One complete out-and-back transfer
- **Red: the height difference** — f: what does not land back

Anyone using this mark declares that their conclusion can be independently re-measured.
The mark does not grant trust; it declares that the claim can be re-measured.

VARIANTS

VARIANTS



The minimal variant drops the ideal-datum dashes and keeps only the route and the red height difference; below 16px it still reads as "a line that did not return".

APPLICATIONS

APPLICATIONS

Benchmark plaque

BM-0
Origin benchmark • AI Origin Community
Annual origin and closure computation
Re-survey records publicly searchable

Reading board • L2 closure stele

closure error f, this cycle
0.26 Tolerance F3 = 0.20
Over F — S08 AI cultural guiding • RT-N
This device is allowed to look bad: publicly displaying failure builds more trust than any success narrative.

Data interface icon

- F1 Tolerance class: the mark's colour
- F2 Tolerance class: the mark's colour
- F3 Tolerance class: the mark's colour

Extension rule

Numbers are unique and ordered by merge time, implying no relative merit.

Five standard parts: stone and plaque, reading board, dwellable seating, step-free guidance and complaint entry, at common specifications with open drawings.

No unlicensed typefaces, images, trademarks or portraits are used.

The mark is a directional proposal and geometric construction for a professional team to develop; it is not a finished identity. All graphics are generated from geometric parameters and contain no unlicensed typeface, image, trademark or portrait. The 0.26 shown on the reading board is illustrative, not a measured result.

FIG.08

a type drawing; it carries no orientation

The chain of custody for innovation elements

ELEMENTS AND THEIR CHAIN OF CUSTODY

Every element answers the same three questions: which space carries it, which benchmark holds its reading, and what gets re-measured.

Element		Spatial carrier		Benchmark		Re-survey item
Land and space	→	Key-area renewal parcels	→	BM-1 / BM-0 / BM-2	→	Delivery rate of promised space
Compute	→	Consolidated compute nodes	→	BM-1	→	Share of public compute open to application
Data	→	Data-factor circulation venue	→	BM-2	→	Authorisation-chain completeness
Scenarios	→	Real-use venues in the two wings	→	BM-21 / BM-22	→	Enforceability of scenario exit
Funding	→	Zhongguancun technology-services wing	→	BM-21	→	Deliberately blank — see the note below
Talent	→	Talent district and third places	→	BM-0	→	Re-survey of service satisfaction

The funding row has no re-survey item. That is a judgement, not an omission.

Investment arrangements are decisions for market actors; an urban design proposal must not turn them into governance metrics, still less into fiscal commitments.
Writing what is not yours to govern into a metric system is its own kind of overreach.

Six global cases, one question

SIX CASES, ONE QUESTION: WHAT MAKES ITS TRUST RE-MEASURABLE?

Id	Case	Trust mechanism	Re-measurable	Transferable point
C1	**Algorithm registers**	Public register of purpose, data, owner and appeal route	High	The information base for a benchmark plaque
C2	**Risk-tiered AI legislation**	Duties differentiated by risk class	Medium	Supports the tiered setting of tolerance F
C3	**Standardised testing frameworks**	Comparable reports from a common toolkit	High	Gives re-survey its technical form
C4	**Algorithmic impact assessment practice**	Mandatory ex-ante and ex-post review for high-risk uses	Med-high	Supports the strictest tolerance for health scenarios
C5	**Civic data-stewardship practice**	Data use decided by a citizen agenda	Medium	Supports public re-survey rights at third-order points
C6	**Open-source reproducibility norms**	Conclusions must ship a re-runnable artifact	High	This package ships a runnable verifier

All six point at the same gap: they register and they assess, but none institutionalises returning to the origin and computing.
A register says what a system declared; an assessment says what experts think. Neither answers how much the conclusion differs when the same public question passes through different nodes at different times.

Case material is drawn from public institutional documents and public reporting. Mechanisms only are cited: no text or images copied, no marks used, no non-public data, and no fabricated company lists, investment figures or output values. Element-to-space correspondence is in geometry/land_use.geojson and public_space.geojson.

LANDMARKS, KIT OF PARTS, SIGNAGE SYNTAX AND OPERATING CYCLE

FIG.09

a type drawing; it carries no orientation

Three AI pilgrimage landmarks

THREE LANDMARKS • spoken in engineering language; no spectacle

L1 The origin benchmark stone

BM-0 • AI Origin Community

Every merged proposal's number is set into the paving — contributors' GitHub IDs are inscribed here. It is not grand. It is accurate, and must be accurate enough to be reused a century later.

L2 The closure stele

BM-1 • Zhongzhiyuan

closure error f, this cycle
0.26 Tolerance F3 = 0.20 **Over F**
S08 AI cultural guiding • RT-N • f = 1 − consistency • illustrative

A public reading wall, updated live, showing each scenario's closure error against its tolerance. Exceedances are marked in datum red. **The value of this landmark is that it is allowed to look bad.**

L3 return-to-datum point

BM-2 • Dazhongsi

An annual civic ceremony space: the line's readings are zeroed here, tolerance revisions are read out, and scenarios returned for re-survey in the past year have their disposition explained. On ordinary days it is a public dwelling space.

Kit of parts for public space: five standard components

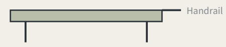
KIT OF PARTS • common specifications and open drawings, so any new node joins in one language

BM-301
Third-order benchmark plaque
Monthly re-survey • unique number

1 Stone and plaque
The number is position, order and cycle

Closure error f
0.14 Level, within tolerance
Next re-survey: March

2 Reading board
Visible on site; no internet needed



3 Dwellable seating
With armrests: older people rely on them to stand

Clear width and gradient enter the re-survey items

4 Step-free guidance
Read by the users themselves

Scan to appeal
or phone / on-site registration
A number on receipt • reply within 5 working days
No number = not accepted; counted

5 Complaint entry
A non-scan method is mandatory, or the right does not exist for P4

Signage numbering syntax (agent.5)

A visitor reading the number knows which order they are standing at and how often it is re-measured

BM-0	**Origin benchmark**	Annual origin and closure computation
BM-1 / BM-2	**First-order benchmark**	Annual re-survey
BM-2x	**2nd-order point**	Quarterly re-survey
BM-3xx	**3rd-order point**	Monthly re-survey
RT-N / RT-S	**Closing route**	Semi-annual re-survey of the whole line

Operations are organised by re-survey cycle, not by festival calendar (agent.6)

Events are therefore governance actions, not publicity

Monthly	**Community day**	third order BM-301/BM-302/BM-303	P4 • P5 • P7
Quarterly	**Scenario open day**	second order BM-21/BM-22	P2 • P3
Semi-annual	**Route re-survey**	first order, routes RT-N / RT-S	P1 • P2
Annual	**Zeroing ceremony**	L3 return-to-datum point	all four review parties present

Developer conversion path: take part in re-survey → propose a scenario → enter the test field → obtain closure clearance → operate
International communication uses published readings as its material, never commitments.

The f=0.26 / f=0.14 on the reading boards are illustrative, not measured results. f is always a deviation: larger is worse, and f ⩽ F is the passing test. Landmark siting requires heritage and engineering review; this proposal does not pre-empt it. The components are a directional proposal for a professional team to develop and are not construction drawings.

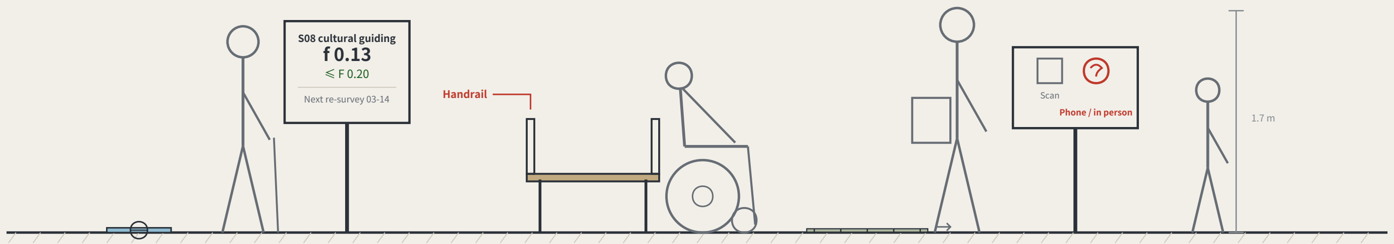
THE BENCHMARK ON THE STREET AND HOW THE KERB IS SHARED

FIG.10

a type drawing; it carries no orientation

What a benchmark looks like at eye level

A BENCHMARK AT EYE LEVEL • third-order BM-301 • elevation 1 m = 138 units



P4 older resident | P5 wheelchair user | P7 frontline worker | P6 child

1 Stone and plaque
Flush with the ground, under a metre across
Uniquely numbered • walk-up

2 Reading board
Visible on site; no internet needed
Exceedance marked in datum red with the date it was returned

3 Dwellable seating
Armrests are mandatory: older people rely on them to stand
A point is only re-measurable if people stay near it

4 Step-free guidance
Continuous clear width and gradient enter the re-survey items
Wheelchair passing is read by the user in person

5 Complaint entry
A non-scan method is mandatory
A number on receipt • reply within 5 working days

Two of the five standard components are hard constraints: seating must have armrests, and the complaint entry must offer a non-scan method. A rule you cannot see in the drawing is a rule the drawing does not contain.

The same kerb, after snow clearance

In winter the order becomes one real trade-off

Piled on the device band Devices stop; S06 falls back to staffed transfer	**Piled on the pedestrian band** Pedestrians pushed into the carriageway - the worst of the three	**Piled on the kerbside parking band** This proposal takes this one The one item in the order that may be lost first

Kerb allocation in priority order

KERB ALLOCATION IN PRIORITY ORDER • sequence and precedence only, no widths

1 Emergency and fire access Never occupied	2 Step-free boarding, turning Not occupied	3 Walking and dwelling Never squeezed below the level-of-service reserve	4 Public transport and cycle parking	5 Device charging and standby Sited only from what remains after the four above	6 Kerbside car parking

satisfied first → → allocated last

The order is itself a position: device charging sits behind pedestrians and accessibility, which means introducing devices must not be paid for with degraded walking conditions. Winter snow storage must be reserved at this stage alongside the device and pedestrian lanes — piled on the device lane the devices stop; piled on the pedestrian lane people are pushed toward the carriageway, which is worse.

The elevation is drawn at 1 m = 138 units. The figures are a scale reference and a statement about who is present, not a rendering. The section expresses sequence and precedence only and carries no widths — capacity must be computed on site from measured clear width, and a figure not derived from measurement is fabricated certainty. Component specifications are in geometry/public_space.geojson and the scenario cards; positions await official boundaries and title verification.

REGIONAL COORDINATION INTERFACES: EXTENDING THE LEVELLING NETWORK

FIG.11

a type drawing; it carries no orientation

An executable form of coordination: how two levelling networks join

AN EXECUTABLE FORM OF COORDINATION: HOW TWO LEVELLING NETWORKS JOIN

This belt's network
Departs BM-0 • F set at BM-1

The adjacent network
Own datum • own F

A joint observation produces a number, not an intention:

Each network reads the same public question set;
the difference is the cross-network closure error. Within a jointly agreed tolerance,
the two networks' records may be mutually recognised; over it, each re-surveys,
and no declaration of partnership substitutes for a reading.

Nobody gives up authority: a joint observation asks only that conventions be comparable, not that organisations merge.

What each of the five partners exchanges, and what it does not

Partner	Stage	Exchanges	Explicitly does not exchange	Precondition for the interface
BDA (Yizhuang)	Volume production and high-level AV road testing	Mutual recognition of closure records across complementary speed domains	Not one admission: an F1 pass here admits a device to pedestrian density, not to a carriageway, and the reverse also holds	Both parties confirm the speed-domain split and test protocols
Future Science City	Corporate research institutes and engineering	Scenario demand and real user density	No claim on IP; no scheduling on anyone's behalf	Wing benchmarks built and a first baseline taken
Huairou Science City	Large facilities and basic research	The measurement method itself: how f is defined, cycle length, the statistics behind revising F	No scenario data is exchanged; the coordination is methodological, not applied	The closure convention published and open to methodological review
Beiwei Community	The innovation community the taskbook names	Reciprocal joint points; exchanged review parties	This proposal does not know its positioning and does not set its benchmark order	Each party names at least one publicly reachable joint point
Beijing-Tianjin-Hebei	Cross-provincial coordination	Cross-provincial recognition of the tolerance convention	No direct recognition of records; without a shared convention they are not comparable	Tolerance definition and revision procedure agreed across provinces

None of the five rows has been confirmed by the party named.
No internal plans are known, nobody is committed, no agreement is assumed; each row is an interface to evaluate independently.

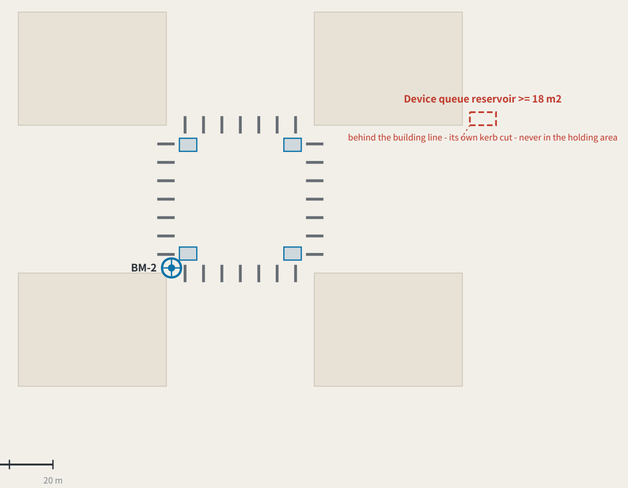
Each partner's stage is described from publicly available general positioning and is unconfirmed by them. The speed-domain relationship in the first row must be settled against both parties' actual test protocols. This sheet proposes interfaces; it commits nobody and is no government determination.

FIG.12

a type drawing; it carries no orientation

I. Four-quadrant plan

FOUR-QUADRANT PLAN • TYPE APPLIED AT BM-2, NOT A SURVEY



Blue blocks are pedestrian holding areas, one per quadrant; the red dashed outline is the device queue reservoir, set behind the building line with its own kerb cut.

If the four quadrants are not connected, devices and pedestrians must contend for the same holding area — which is why this district names it the most concrete spatial task it has.

2. Dimensions: design intent / as-measured

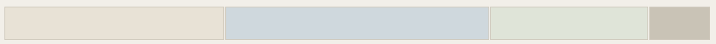
DIMENSIONS: DESIGN INTENT vs AS-MEASURED

Item	Design intent	As measured
Pedestrian holding area, per quadrant width >= 3.0 m x depth >= 4.0 m	⩾ 12 m²	
Crossing effective clear width after deducting poles, covers and the furnishing strip	⩾ 4.0 m	
Kerb ramp gradient Barrier-Free Environment Construction Law and current codes	⩽ 1:12	
device queue reservoir set behind the building line, with its own kerb cut	⩾ 18 m²	
Reservoir to holding area, clear distance must not cross any desire line	⩾ 2.0 m	
BM-2 stone to pedestrian edge the stone is flush with the ground and is not a trip hazard	⩾ 0.6 m	

The right-hand column is deliberately empty.
Capacity is computed on site from measured clear width. A number filled inwithout measuring is manufactured certainty. Completed before any device is admitted.

3. Kerb allocation section (order and intended widths only)

KERB ALLOCATION SECTION



(1) Frontage	(2) Through walking	(3) Furnishing strip (BM-2 sits here)	(4) Kerb
2.5 m	3.0 m	1.8 m	0.7 m

Building line to kerb: frontage, walking, furnishing, kerb.BM-2 sits in the furnishing strip, so a reading is never taken in the walking zone.

This is a type drawing applied at BM-2, not a survey of the existing junction. Every dimension is design intent and the as-measured column is deliberately empty: carrying capacity must be computed on site from measured effective clear width. Junction geometry, channelisation and signals rest with the road authority and a traffic study.

The three key areas in section, at one scale

FIG.13

One horizontal scale • 220 m across each • no vertical exaggeration • ±0.00 is that area's own benchmark, not a shared absolute datum

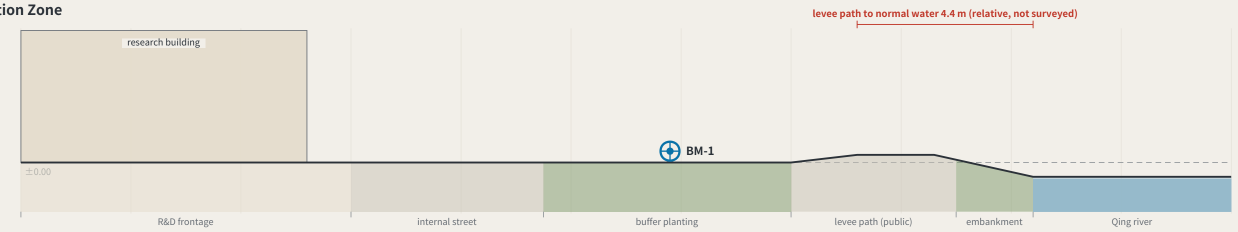
a type drawing; it carries no orientation

Zhongzhiyuan AI Acceleration Zone

±0.00 = BM-1

192.9 ha • BM-1
dominant use R&D 100%
2 land-use classes inside

External arrival. The area is 100% single-use, so public access to the Qing river runs through the block or not at all — what this section sets out is a public route across the R&D frontage and onto the levee path.



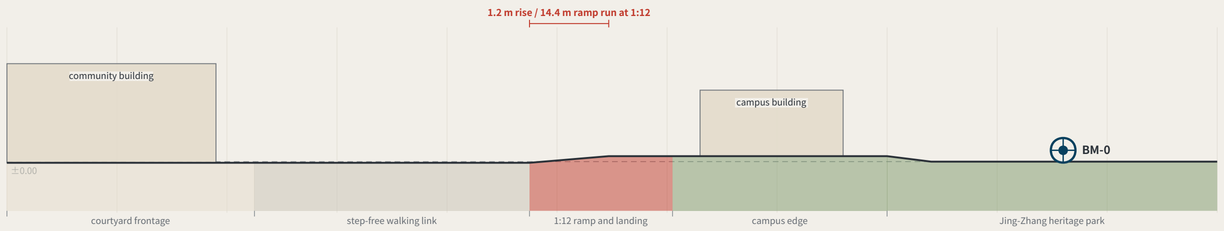
No flood level is drawn. The design water level is official data this package does not hold, and drawing an elevation with no source behind it is the defect shape this package's own errata register counts.

Beijing AI Origin Community

±0.00 = BM-0

104.3 ha • BM-0
dominant use community services 94%
2 land-use classes inside

A 1.2 m grade change, and step-free continuity. The campus edge ends in steps, and P5 needs a continuous step-free route. The reason this section exists is to put a 1:12 ramp with a landing where the steps are.



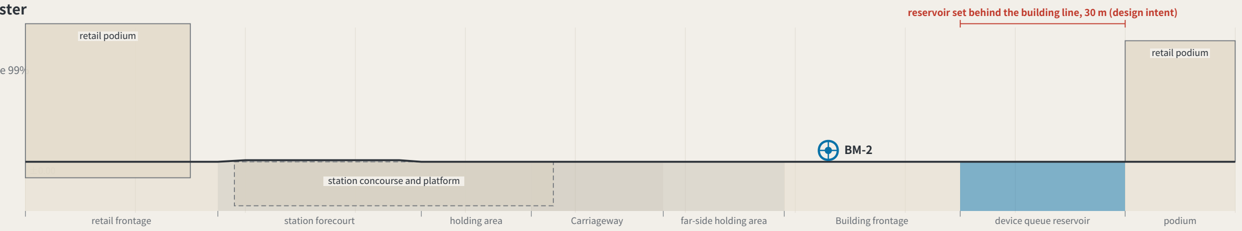
Walking the 220 m: typical adult 3.1 min • P4 4.1 min • P5 step-free 6.1 min (speeds are assumptions, not measurements)

Dazhongsi AI Industry Cluster

±0.00 = BM-2

72.0 ha • BM-2
dominant use industry and commerce 99%
2 land-use classes inside

A station below grade and a queue behind the building line. The only section here with anything below ground, and the only one with a carriageway. The device queue reservoir sits behind the building line with its own kerb cut, clear of the crossing desire lines — the four-quadrant plan is FIG.12.



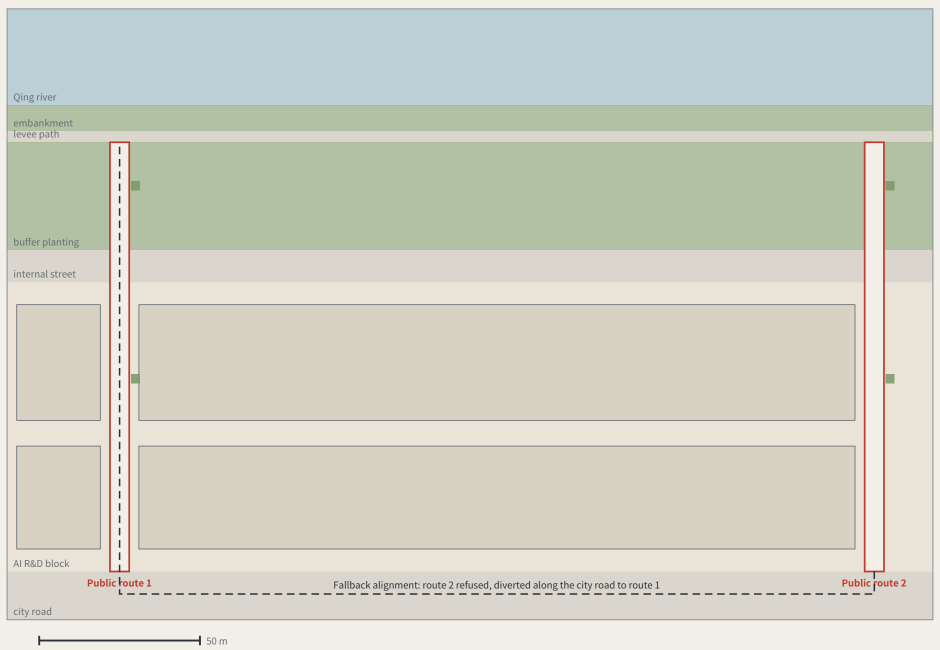
One horizontal scale, one datum convention, no vertical exaggeration. They are placed together so the charge that the three are one template used three times can be tested: one reaches a river across a 100% single-use block, one is step-free continuity over a 1.2 m rise, one is a station below grade with the device queue behind the building line. ±0.00 is each area's own benchmark — absolute elevations need official levelling data this package does not hold, so they are not drawn. Concept design intent, not surveyed profiles.

THE LEVELING LINE • Haidian, Beijing • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-17 (sources.json; 1 not obtained) Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

The through-block public route to the Qing river

FIG.14

a type drawing; it carries no orientation



Dimensions: design intent and as built

The right column is deliberately empty — effective clear width is filled in on site

Item	Design intent	As measured
Public route clear width two abreast, and a wheelchair passing	⩾ 6.0 m	
Route centre spacing a 939.0 m frontage needs 4 routes, 234.8 m apart	⩽ 250 m	
Longitudinal gradient 133.5 m rising 1.4 m, an average of 1:95, step-free throughout	no stretch steeper than 1:20	
Opening hours a route that can be locked is not public access; it is permission	24 h, ungated	
Spacing of places to sit standard component KIT-03, with armrests	⩽ 60 m	
Levee path clear width two-way walking, passing	⩾ 3.5 m	
Building setback from the route the ground floor may not face it with a solid wall	⩾ 3.0 m	

The count is not a preference. It is a division.

Under the provisional substitute boundary the northern frontage of Zhongzhiyuan runs 939.0 m. At a maximum centre spacing of 250 m that is 4 public routes, 234.8 m apart. Replace the boundary with an official polygon and the number moves with it — it is computed from the geometry at build time, not written onto the sheet.

What one gate costs

With the route open it is 133.5 m from the city road to the levee path. With route 2 refused and the walk diverted along the city road to route 1, it is 368.3 m. For P0.6, walking at 5 m/s, 3.7 minutes becomes 10.2 minutes. That is the working unit of a public route that can be locked.

The route along its length: city road to levee path

2.6× the scale of the plan • no vertical exaggeration • 133.5 m rising 1.4 m, an average of 1:95

The steepest stretch is inside the planting, and 1:20 is the constraint on that stretch — an average gradient would hide it.

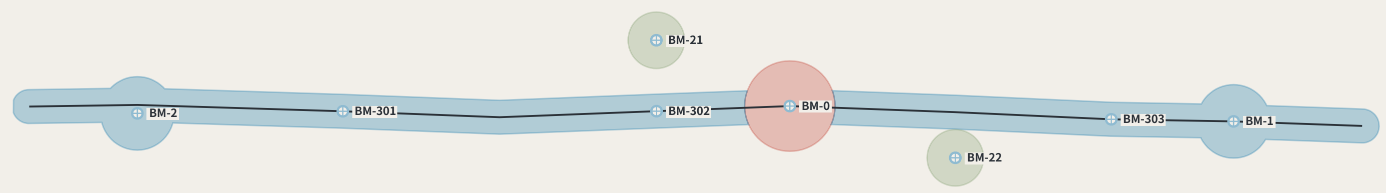
FIG.13's section says this edge needs a public route across the R&D block onto the levee path, and no drawing showed it. This is that route: 6.0 m clear, centres at most 250 m apart (939.0 of them across a 4 m frontage), open 24 hours and ungated. The fallback alignment and the cost of a refusal are drawn. A type drawing, not a survey; dimensions are design intent, the as-measured column is empty; boundaries are provisional.

THE LEVELING LINE • Haidian, Beijing • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-17 (sources.json; 1 not obtained) Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

Phasing: advanced by closure results, not by dates

FIG.15

north is right (unrolled)



Phases advance on closure results, not on years

Phase	Increment	Spine added	Benchmarks this phase adds	Condition that opens it (not a date)	Re-surveys/yr	Annual running cost
Near	32.1 ha	0.64 km	BM-0	none — no prior closure needed	1	**CNY 2,484–7,922**
Mid	234.8 ha	8.80 km	BM-1, BM-2, BM-301, BM-302, BM-303	four near-term items done; f ⩽ F ×2	39	**CNY 13,864–45,492**
Long	25.1 ha	—	BM-21, BM-22	mid phase: f ⩽ F ×2	47	**CNY 19,444–63,922**

The table above phases the eight points that exist and their cumulative cost. FIG.21 shows that meeting the fifteen-minute rule needs 9 more third-order points (a further CNY 22,005–76,358 a year), falling by order into the mid and long phases — this is not the phasing of a network that already complies.

The first phase is a stone, a plate and one reading a year

The near-term increment is 32.1 ha and holds BM-0 alone: 1 re-survey a year, CNY 2,484–7,922 to run. That line is the answer to "this mechanism is too heavy" — it can start from one point, and that point costs what a sub-district office can approve by itself.

The condition is a number, not a year

The mid phase does not begin because a second year arrived. It begins because the four near-term items are complete and f ⩽ F for two consecutive cycles; the long phase the same. Once may be luck — the same rule this proposal writes at PATH-4 of the conversion path, applied to phasing. Fail it and the programme stays where it is, which makes the phasing table an exit table as well.

Which four are the four near-term items

The condition that opens the mid phase reads "the four near-term items complete" and nothing in the package says which four — and the proposal contains two different sets of four: the near-term projects R1–R4, and the ones needing no official data, R1–R3 and R8. The second set contains a long-term item, so reading the gate against it makes the gate circular. Taking the first:

R1 datum stone L1 and the public evidence hall — one complete closure and its reading published within the first cycle
R2 first public setting of the tolerance F — a first version of F1/F2/F3 each, posted
R3 the four-week closure trial on cultural guidance S08 — closure recomputed and the consistency ratio published within four weeks
R4 third-order benchmarks set out at communities and rail stations — one recorded reading at each of the three community points

Two naming errors this sheet found

Dropping each benchmark into the phase polygons contradicts two of the phase names. The mid phase was named for "three first-order points" and contains two (BM-1, BM-2); the third is the datum BM-0, which is in the near phase. The long phase was named for "the full three-order network", and the three third-order community points already fall inside the mid phase — what the long phase actually adds is the two second-order wing access points. The names were written for the intent and never checked against the polygons. Both are corrected in the layer, and every cell of this table is computed from the geometry rather than typed.

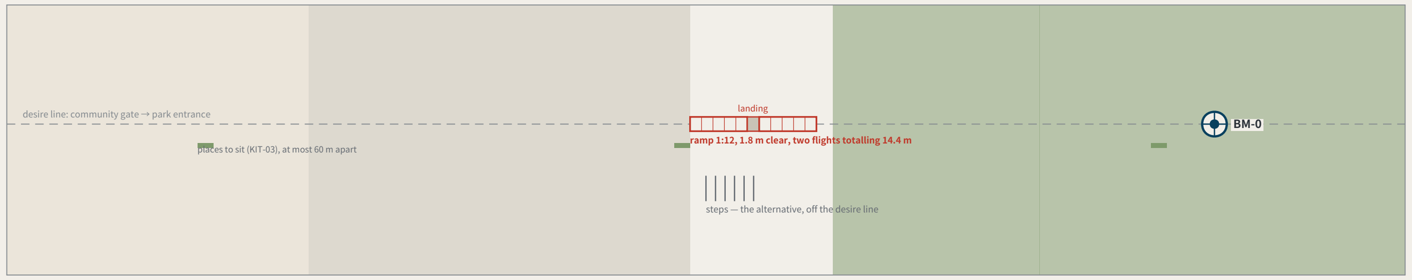
phasing.geojson has shipped for a long time, been repaired once and enters the recompute path, and no drawing had ever shown it. This sheet draws the three increments and costs each phase's year with the same model as build_operations, not a second set of coefficients. Phases open on closure results, not on years. The geometry is concept advice on provisional boundaries and must be recomputed when official ones are published.

THE LEVELING LINE • Haidian, Beijing • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-17 (sources.json; 1 not obtained) Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

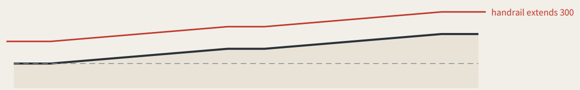
FIG.17

a type drawing; it carries no orientation

I. Plan



II. The ramp along its length (no vertical exaggeration)



rising 1.2 m in two flights of 7.2 m with a 1.5 m landing between

5 m

What offsetting the ramp would cost

The usual arrangement puts the steps on the desire line and offsets the ramp. Offset it by 22 m and whoever uses it walks 44 m further every time — out and back. For P0.6, at 5 m/s, that is 1.2 minutes a trip; at twice a week for a year, 4.6 km. Nobody on the steps walks a metre further for the ramp being on the line: steps are shorter, so they sit beside it.

Walking the 176 m

typical adult	**2.4 min**
P4	**3.3 min**
P5	**4.9 min**

speeds are assumptions, not measurements

Dimensions: design intent and as built

The right column is deliberately empty — as-built values are measured on site

Item	Design intent	Why this number	As measured
Ramp gradient	**1:12**	step-free throughout; the ramp is on the desire line and the steps are its alternative	
Ramp clear width	**⩾ 1.8 m**	two people passing, or a wheelchair with someone alongside	
landing	**one every 750 mm of rise, at least 1.5 m long**	a 1.2 m rise split into two flights — there has to be somewhere to stop	
Handrail	**both sides, extending at least 300 mm past each end**	a handrail that stops at the top of the slope stops where it is still needed	
Level approach at each end	**at least 1.5 m level**	level ground before and after, so nobody has to turn on the slope	
Spacing of places to sit	**⩽ 60 m**	standard component KIT-03, 450 mm high with armrests	
Steps	**beside the ramp, off the desire line**	anyone who prefers steps is not sent around — steps are shorter than the ramp	

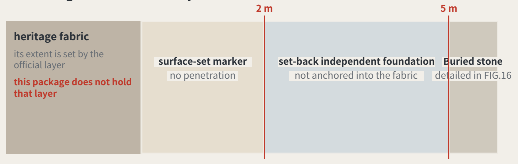
FIG.13’ s section says this place exists to put a 1:12 ramp where the steps are, and no drawing in the package had shown it. One design decision: the ramp sits on the desire line and the steps beside it — offset the ramp by 22 m and its user walks 44 m further every trip, while nobody on the steps walks a metre more. A type drawing, not a survey; dimensions are design intent and the as-built column is empty.

Monuments beside heritage fabric: the no-drilling construction and its setback rule

FIG.19

a type drawing; it carries no orientation

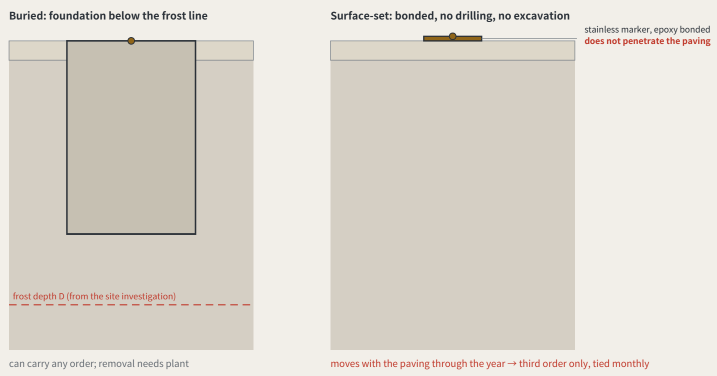
I. Choosing the construction by distance from the fabric



d = horizontal distance to the outer edge of the fabric

2 m

II. Two constructions, one scale



The rule is complete; its input is missing

Which line d is measured from depends on the officially published heritage protection area and its construction-control zone. This package does not hold that layer: each of the 3 shipped constraint features records it as a data gap in its own note, and this proposal does not infer it. So the sheet gives the rule, the thresholds and the three constructions, and leaves that line to whoever publishes it — guessing a protection boundary to make the drawing look complete would mean inventing the one thing here that is not this proposal’ s to decide.

III. What each construction costs

Construction	Applies when	Foundation	Orders it may carry	Re-survey cycle	What removal leaves
Buried stone	d ⩾ 5 m	below the standard frost depth D + 100 mm	datum / first / second / third	per order	plant to lift out; leaves an excavation
set-back independent foundation	2 ⩽ d < 5 m	below frost line, set back ⩾ 2 m, unanchored	second / third	per order	plant to lift out; leaves an excavation
Surface-set marker (no penetration)	d < 2 m	no foundation; bonded to existing paving, no drilling	**third order only**	**monthly, and tied to the nearest buried point each time**	lifting it leaves only a bond mark, which cleans off

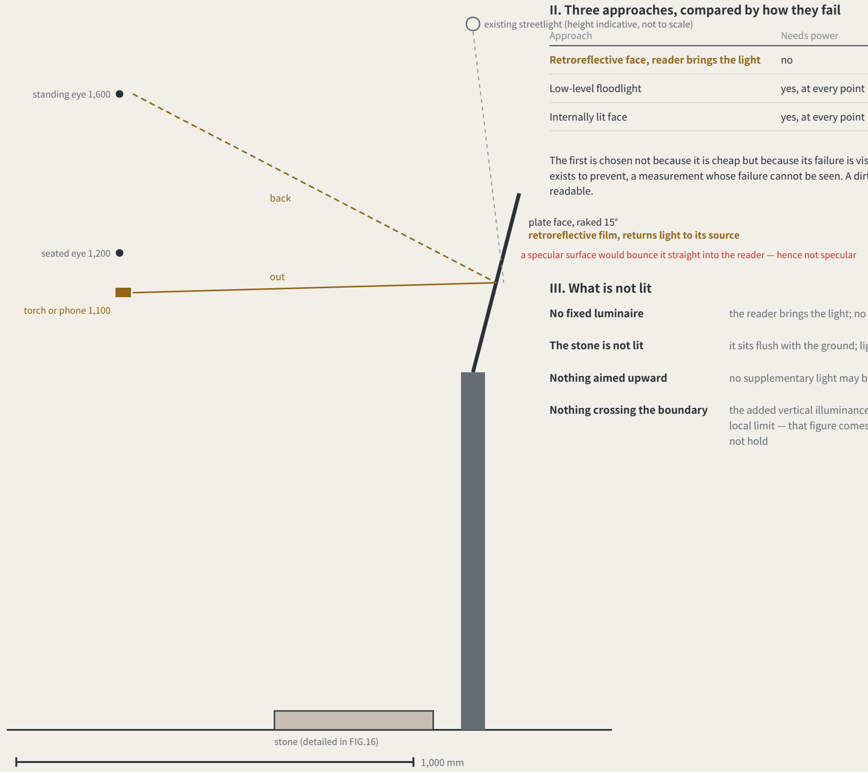
The call is for a belt along a hundred-year-old railway; this proposal’ s central act is sinking concrete. Three constructions, chosen by distance from the fabric, with the cost stated: without excavation there is no foundation below the frost line, so the surface-set marker may not carry first- or second-order duty. Where d is measured from depends on the official heritage layer, absent here and recorded as a gap in constraints.geojson.

Reading after dark, without lighting the benchmark

FIG.20

a type drawing; it carries no orientation

I. Geometry: where the light comes from, and returns to



II. Three approaches, compared by how they fail

Approach	Needs power	How it fails	Light trespass
Retroreflective face, reader brings the light	no	the face gets dirty → visible in daylight → caught before it stops being readable	none
Low-level floodlight	yes, at every point	the lamp dies → the plate looks intact and cannot be read	yes, needs controlling
Internally lit face	yes, at every point	the lamp dies → the whole face goes, and it still looks like a plate	yes, and aimed at the reader

The first is chosen not because it is cheap but because its failure is visible. When a lamp dies the plate still looks intact and can no longer be read — exactly the thing this proposal exists to prevent, a measurement whose failure cannot be seen. A dirty retroreflective face is simply dirty, visible in daylight to anyone walking past, well before it stops being readable.

III. What is not lit

No fixed luminaire	the reader brings the light; no power supply, no luminaire to maintain
The stone is not lit	it sits flush with the ground; lighting it adds no legibility, only night-time glare
Nothing aimed upward	no supplementary light may be emitted above the horizontal
Nothing crossing the boundary	the added vertical illuminance at neighbouring dwelling windows must meet the local limit — that figure comes from the local standard, which this package does not hold

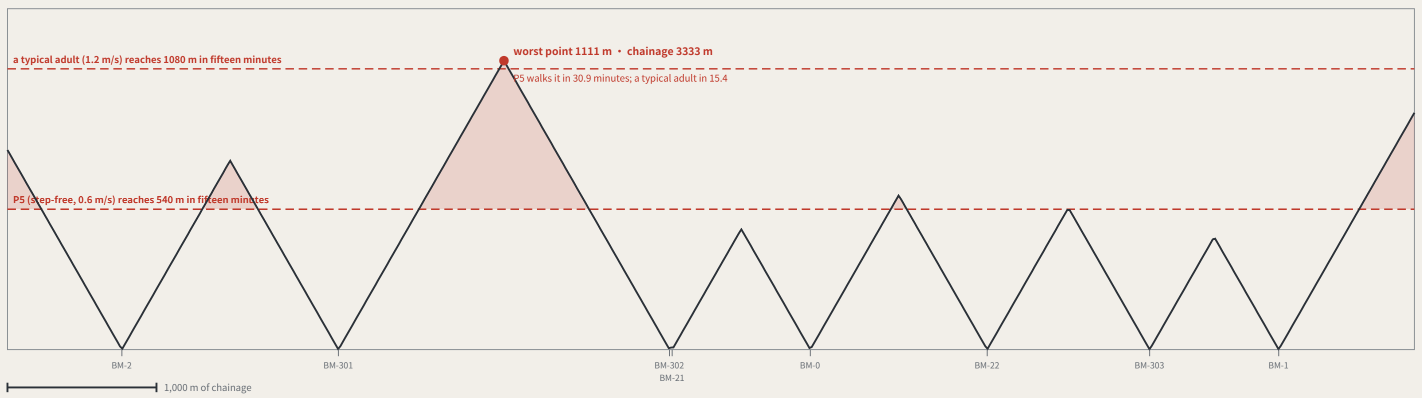
Three third-order points are read monthly by residents, and much of the year those days fall after dark — and the package had never said how the plate is read at night. The obvious answer is a lamp, and it is the wrong one: it needs power, needs maintaining, throws light into the windows opposite, and when it dies the plate looks intact and cannot be read — a measurement whose failure is invisible, which is the thing this proposal exists to prevent. So the face is retroreflective with no fixed luminaire and the reader brings the light; a dirty face is simply dirty, in daylight.

How far the nearest benchmark actually is: this proposal's own rule, applied to itself

FIG.21

a type drawing; it carries no orientation

I. Distance to the nearest benchmark, along the spine



1,000 m of chainage

II. Turning that sentence into a spacing rule that can be checked

Where the rule comes from	this proposal's own repeated claim: a right of review you must walk fifteen minutes to reach has not been given
Interior spacing	chainage between adjacent points ⩽ 2 × 540 m = 1080 m
Both ends of the line	end of line to its nearest point ⩽ 540 m
Which speed the rule uses	P5's 0.6 m/s, not a typical adult's 1.2 m/s — setting the rule by the faster figure sets it at a standard most people cannot use

III. How far short the current layout falls

Six of the nine segments between the eight existing points fail. Closing the gap needs **9 more third-order points** (or 3 if the rule is only held to an average walker). They are not free: third-order points are read monthly, and every one of them enters the annual operating table. **This sheet does not place them** — putting 9 coordinates on a provisional substitute spine is the manufactured certainty this register counts. The rule ships; the positions wait for the official alignment.

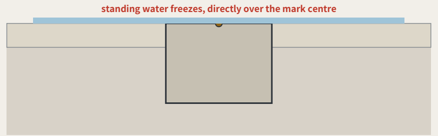
This proposal argues repeatedly that a right of review you must walk fifteen minutes to reach has not been given, and its entire participation argument rests on anyone being able to walk up to a benchmark and take a reading — and nobody had ever measured that walk. Measured: the worst place on the line is 1111 m from the nearest benchmark, 5 minutes for P30.9, twice the limit this proposal holds others to. The rule ships as a spacing requirement; the positions wait for the official alignment.

FIG.22

a type drawing; it carries no orientation

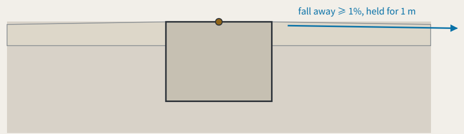
I. Two ways of doing it, at one scale

Wrong: level paving, no fall



Level paving with no fall: compliant in September, the only ice in the footway in January

Right: the stone as a local high point



The stone is the local high point: flush with the paving beside it, and that paving falls away

1,000 mm

The fall is drawn true: 1% over 1 m is 10 mm, which is barely visible at this scale — and a slope nobody can see on a drawing is a slope nobody checks on site, while every dimension on the sheet is still satisfied.

II. What that requires of the construction

Item	Requirement	Why
Stone top against the paving	**flush, tolerance ±5 mm**	KIT-01: no trip hazard
Fall around it	**⩾ 1%, away from the stone, held for ⩾ 1 m**	makes the stone a local high point — water leaves instead of gathering
Protective ring	**none**	a recessed ring is a bowl; a raised one is the trip hazard KIT-01 forbids
Surfacing	**permeable or tight-jointed; no joint channel running inward**	a joint channel leads water to the mark centre
Winter reading	**taken after clearing, with the condition recorded**	a cleared reading and an uncleared one are not directly comparable

III. Three things a winter reading adds

Clear it — clear snow and ice from the stone and 300 mm around it before reading
a reading taken over a film of ice is measuring the ice

Check the drainage — confirm the stone is still the high point: no ponding marks or ice within 1 m
a ponding mark is evidence the drainage has already failed, and it appears before the ice

Record the condition — mark the cycle record as a winter condition, and whether the stone was cleared
cleared and uncleared readings are not directly comparable — unrecorded, nobody knows whether they should be

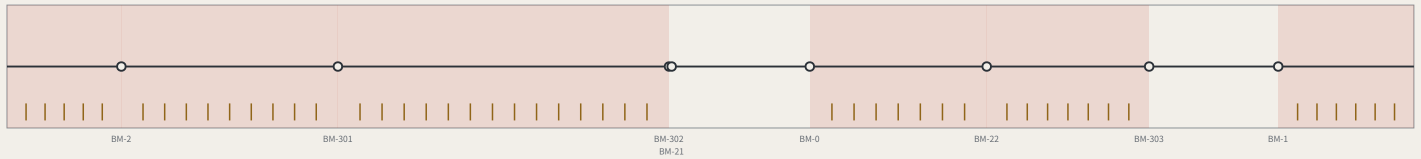
The three third-order points are read monthly, January included, and this package had never said what that costs. KIT-01 requires the stone flush with no trip hazard and FIG.16 gave that ±5 mm — but a mark flush in level paving sits in standing water: the tolerance that makes it safe in September makes it dangerous in January. The answer is a local high point, not a ring.

Finding the nearest benchmark: which way, and how far

FIG.23

north is right (unrolled)

I. Markers only where this proposal already breaks its own rule



Red marks the stretches FIG.21 finds short of the fifteen-minute rule; one marker every 150 m, 48 in all. Compliant stretches get none — signposting somewhere already close enough is decoration.

1,000 m

II. Four things the face has to answer

← BM-302
410 m • P5 about 11 min

BM-0 →
520 m • P5 about 14 min

standard component KIT-04; nothing new is built

Direction — **← BM-302 / BM-0 →**
on a line either end may be nearer; only at the midpoint are they equal

Distance — **410 m / 520 m**
metres, because whoever walks it can check metres

Time — **P5 about 11 min / 14 min**
at 0.6 m/s, the same figure every other sheet here uses (personas.json)

Id — **BM-302 • BM-0**
the reading taken on arrival is attributed by it, without asking anyone where they stood

III. What signage cannot do

These 48 markers do not make 1111 m near. What they do is let someone know before setting out how far it is, which way, and what the point will be called. That is mitigation, not the fix. The fix is the 21 third-order points on FIG.9, whose positions wait for the official alignment. Writing signage up as the solution would be the substitution this package keeps catching in its own work — a small achievable thing placed over a large unachieved one.

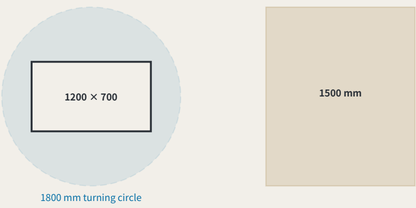
FIG.21 measured the worst walk on the line at 1111 m, with six of nine segments short of this proposal's own fifteen-minute rule. Closing it needs 9 third-order points whose positions wait for the official alignment. People still have to find the ones that exist. Hence markers — only inside the failing stretches, one every 150 m, 48 in all, on the existing KIT-04 so nothing new is built. Signage does not make 1111 m near: this is mitigation, not the fix.

The device envelope: what exactly goes in those 18 m²

FIG.24

a type drawing; it carries no orientation

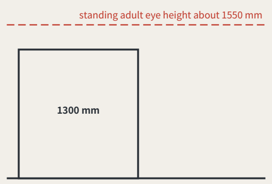
I. The envelope and its turning circle, in plan



The accessible clear width. Nothing a device does — passing, parking or queueing — may push it below this. The pedestrian space is set first and the device gets what is left (FIG.12).

1,000 mm

II. Elevation: why the height is that number



The height sits below eye level so a queueing device does not cut the sightline across the footway. **Nothing outside this package fixes it; this proposal is choosing it** — eye height varies by person, and what the sheet gives is the reasoning for the value, not a standard to copy.

III. Each dimension, and what bounds it

Item	Value	What bounds it	Why this number
Length × width	**1200 × 700 mm**	**already published**	Derived from the separate kerb cut in FIG.12: a device that cannot pass the cut is not admitted here
Height, loaded	**1300 mm**	**chosen here**	Set below standing eye height so queueing devices do not block the sightline across the footway; nothing outside this package fixes it
Turning circle	**1800 mm**	**chosen here**	What a standing turn needs. It fixes the shape of the reservoir and not merely its area — the half that "18 m²" could not say
Kerb mass	**⩽ 120 kg**	**chosen here**	The order of magnitude one person can push clear after a power failure; above it the kerb cut has nowhere to put the machinery that would be needed
Speed on site	**⩽ 6 km/h**	**chosen here**	The speed bound for shared pedestrian and vehicle space, taken at the order of walking pace; no citable local rule was obtained, so it is registered as a value chosen here
Clear width left after passing	**⩾ 1500 mm**	**already published**	The accessible clear width, the same figure as KIT-04; a device passing may not reduce it below that
Marking	**⩾ 25 mm cap height**	**already published**	id, responsible party and appeal route, all three; the same rule as the reading plate — attribution needs a legible id (FIG.16, FIG.20)

IV. Turning 18 m² into a device count — the two sheets have to agree

One standing bay is the envelope length times the turning circle: 2.16 m². FIG.12 publishes a 18 m² floor for the reservoir, so that is **8 at once**. The number did not exist before: "⩾ 18 m²" never said how many, and two devices of equal footprint with different turning radii do not need the same reservoir. The build checks the two sheets against each other — if the area does not divide into bays, or the count exceeds what the floor holds, it refuses.

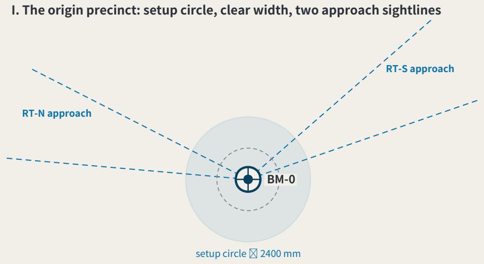
FIG.12 fixed the device reservoir at ⩾ 18 m² and this package had never drawn a device. Area is not specification: equal footprints with different turning radii need different reservoirs. An envelope, not a product; 7 dimensions each marked with what bounds them, 4 bounded by nothing but this proposal — **attack those first**. It holds 18 m² / 8 devices.

BM-0, the origin: the place both routes have to return to

FIG.25

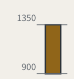
north is up

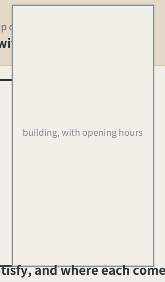
I. The origin precinct: setup circle, clear width, two approach sightlines



accessible clear width 1500 mm — the setup c

II. The closure record, readable wi





A reading is published the moment it is taken, and evidence is only as public as the door in front of it. So the closure record sits **outside**, facing the footway, on the reading-plate geometry FIG.16 already fixed (900–1350 mm to the face). Hang it in a hall and the sentence "anyone can walk up to a benchmark and take a reading" quietly becomes "during opening hours".

III. Six things the origin has to satisfy, and where each comes from

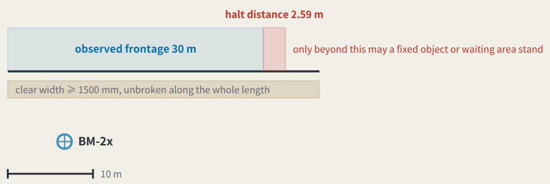
Item	Requirement	Why, and where from
Setup circle	**⌀ 2400 mm**	a tripod spreads to 1,200 mm and the operator needs 600 mm outside it; closing a route means standing an instrument on this stone, and this circle is what that actually takes
Against the clear width	**⩾ 1500 mm still left outside the setup circle**	setting up may not eat the clear width — the same rule FIG.24 applies to devices, and this package does not get a looser one for its own surveyors
Sightlines for both routes	**one approach cone each, no trees planted and nothing parked inside it**	RT-N and RT-S terminate here, and terminating means seeing back to the last turning point. Blocked, you add a turning point — and every added turning point is one more height difference that can go wrong
Which way the record faces	**facing the footway, 900–1350 mm to the face, readable without going inside**	a reading is published as it is taken, and evidence is only as public as the door — an origin with opening hours is shut at the moment the reading lands
What it displays	**network closure error**	Network closure only. A point' s own failure is shown at that point (errata E50); moving all the bad news back to the origin is precisely the defect this register recorded
Cross-fall	**⩾ 1% falling away**	as FIG.22: the stone has to be a local high point, or in January it is the one patch of ice here — and the instrument stands on it too

BM-0 appears on nearly every sheet here and had never been drawn — and what it carries is a spatial property: **a reading is published the moment it is taken, and evidence is only as public as the door.** Hang the record in a hall with opening hours and the origin is shut at the moment the reading lands. This sheet gives the setup circle, an approach sightline cone for each closing route, and the closure record outside the door facing the footway.

FIG.26

north is right (unrolled)

I. The observed frontage and the halt distance, at one scale



II. This distance is computed, not decided

FIG.24 fixes site speed at ⩽ 6 km/h = 1.667 m/s. Halt distance = reaction + braking = $1.667 \times 1 + 1.667^2 \div (2 \times 1.5) = 1.67 + 0.93 =$ **2.59 m**. **The reaction time and the deceleration are chosen by this proposal and are not standards** — registered as `A-DEVICE-002` on the register' s own terms, with derivation, consequence if wrong, closure condition and owner, in the same commit as this sheet. The formula ships with the drawing: substitute your own figures and this ground recomputes.

III. What makes second order differ from third

Item	Value	Why
The stone	KIT-01, the same as a third-order point	second and third order differ in cadence and route role, not in construction — a more important place does not get a special benchmark
Re-survey cycle	quarterly	the wings are where measurement density comes from; readings are quarterly, priced in the same table as first-order annual and third-order monthly
Observed frontage	30 m	the length of kerb along which a scenario is actually observed. "Scenarios take their readings here" with no length is the same kind of sentence as "18 m² of reservoir" with no device in it
Halt distance	2.59 m	computed from the site speed FIG.24 fixes, ⩽ 6 km/h: 1 s of reaction covers 1.67 m, then decelerating at 1.5 m/s² covers 0.93 m
Clearance required	no fixed object and no waiting area within 2.59 m ahead of the frontage	a scenario that cannot stop is not a scenario but an incident; this ground is the physical form of the sentence "it can be stopped"
Clear width	⩾ 1500 mm, unbroken	the same figure as FIG.24 and FIG.25 — observing, setting up an instrument and a device passing may none of them eat it

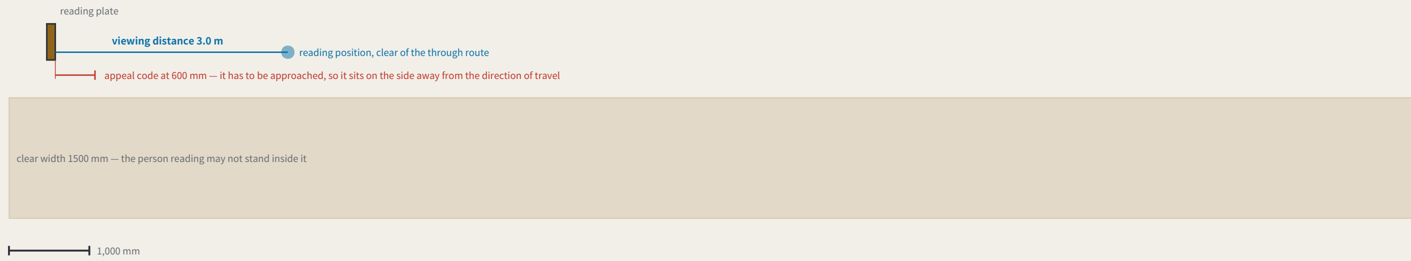
The two wings carry the second order — in the naming table, the cost table and the resource table — and had never been drawn. What differs from third order is not the stone (KIT-01, as everywhere) but that what runs here is **a scenario against real users**, so it has to be stoppable. From FIG.24' s ⩽ 6 km/h the halt distance is 2.59 m, and no fixed object or waiting area may stand in that ground ahead of the frontage. **The reaction time and deceleration are chosen by this proposal and registered as A-DEVICE-002.**

One reading: where a person stands, and whether they block the way

FIG.27

a type drawing; it carries no orientation

I. Where the person reading stands



II. This viewing distance comes from a ratio this package already published

This package has long used "legible height ≈ viewing distance ÷ 250" to check its own drawings, finding a minimum actual cap height of 4.73 mm, all passing. **The same ratio had never been pointed at the plate.** Pointed there, what it yields is not a type size but **a place to stand**: a 12 mm cap height gives 3.0 m, and 3.0 m is outside the 1500 mm clear route — the reader and the people walking past neither yield nor collide. At 6 mm the distance falls to 1.5 m and the reader stands in the middle of the path: **the plate is still legible and the footway no longer works**. The 12 mm cap height is chosen here and registered as `A-PLATE-001`. **This package fixes another cap height too**: FIG.24 requires a device marking at ⩾ 25 mm, legible on the same ratio from 6.25 m. That is neither a contradiction nor an accident — a device marking identifies a machine **coming toward you**, and has to be read while there is still time to act; a reading plate is read standing in front of it. Different distances, different type. The relationship is stated because two cap heights with none stated between them is how a reader concludes one of them was picked carelessly. The ratio of 250 is not chosen here — it is the one this package already holds itself to.

III. The four steps of one reading, and the ground each takes

Step	What happens	Why that dimension
Walk there	FIG.21 measured the worst walk on the line at 1,111 m, 5 minutes for P30.9 — this proposal does not satisfy its own rule	the cost of this step has been measured, and it came out failing
Stand	read from 3.0 m, standing clear of the through route	12 mm cap height × the 250 ratio = 3.0 m; at 6 mm it falls to 1.5 m and the reader stands in the middle of the path
Read	read the figures on the plate and the benchmark_id	the id lets the reading be attributed without asking anyone where they were standing
Disagree	scan the appeal code, approaching to about 600 mm	this step has to come close, so the code sits on the side away from the direction of travel — the one action that must intrude does so at the edge rather than across the flow

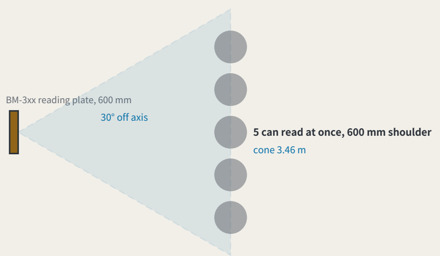
The whole participation argument rests on one sentence — anyone can walk up to a benchmark and take a reading — and no sheet had drawn the four minutes that person spends standing there. Standing is not free: the ground they occupy is the ground everyone else walks through. Working back from the "legible height ≈ viewing distance ÷ 250" ratio this package already publishes, a 12 mm cap height gives 3.0 m, falling outside the 1500 mm clear route; at 6 mm the reader stands in the path. The cap height is chosen here and registered as A-PLATE-001.

Third order: the most numerous tier, and more than one reader

FIG.28

north is up

I. The legible cone, and how many can read at once



II. The part ground cannot solve

A class of 30 cannot read at once. 5 fit the cone, so 6 rounds, and at 40 s a reading that is about 4 minutes. **The number is here because widening the ground does not solve it.** No cone holds thirty people; the rest take turns, and turns cost time — a cost that, unwritten, gets discovered on site in the form of "this point is awkward". The off-axis angle and the shoulder width are chosen by this proposal and registered as `A-READ-001`; the plate width, the viewing distance and the clear width come from FIG.16, FIG.27 and FIG.24 and are not fixed here.

clear width 1500 mm — a queue runs parallel to the kerb, never pushing into this

1,000 mm

III. Standing rules for a third-order point

Item	Value	Why
Re-survey cycle	monthly	third order is the most numerous tier and the one residents actually stand at
Legible cone	3.46 m	a 600 mm plate (FIG.16) stays legible within 30° off axis, and at the distance FIG.27 fixes, 3.0 m, the cone opens to this width
Readers at once	5	cone ÷ a 600 mm shoulder. **This is this sheet answering the question FIG.27 left open** — that sheet fixed one person' s position
A class of thirty	6 rounds, about 4 minutes	past that count the extra people are not handled by ground but by taking turns — **and turns are a time cost, which belongs in writing rather than being discovered on site**
Which way a queue runs	parallel to the kerb, never perpendicular to the direction of travel	a perpendicular queue pushes its tail into the 1500 mm clear width; a parallel one does not
What is not provided	no railing, no dedicated waiting area	a third-order point sits in a neighbourhood, and fencing it stops it being something you can read in passing; what this sheet gives is a standing rule, not a new structure

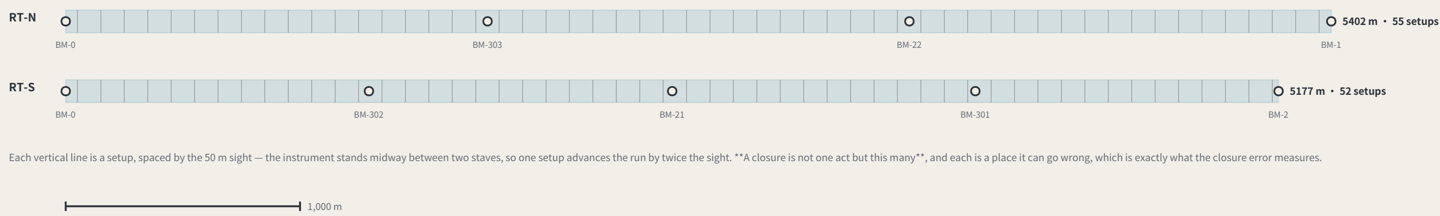
The origin and the second order each have a sheet; **the third order is the tier nobody drew, and the one residents actually stand at**. It also inherits what FIG.27 left open: that sheet' s 3.0 m distance is **one person' s** position. From a 600 mm plate and a 30° off-axis limit the cone is 3.46 m wide and 5 people read at once; a class of 30 takes 6 rounds, about 4 minutes — **that part is solved by turns, not by ground**.

The annual first-order closure: how long a route actually takes

FIG.29

north is right (unrolled)

I. The two closing routes, and how many setups each takes



Each vertical line is a setup, spaced by the 50 m sight — the instrument stands midway between two staves, so one setup advances the run by twice the sight. **A closure is not one act but this many**, and each is a place it can go wrong, which is exactly what the closure error measures.

1,000 m

II. The duration derived from geometry, against the band the cost table publishes

Route	Length	Setups	Observing	Walk back	Total
RT-N	5402 m	55	4.6 h	1.3 h	5.8 h
RT-S	5177 m	52	4.3 h	1.2 h	5.5 h

the band the cost table publishes for a first-order session: 4.0–6.0 h

III. These two numbers had never met

The 4.0–6.0 h for a first-order point in `operations.json` was written into the model by hand, on its author' s judgement, and nothing could check it. The route lengths are not hand-written: they come out of the shipped geometry. Divide by the sight into setups, multiply by the minutes a setup takes, add the walk back along the spine, and the result is 5.5–5.8 h — **inside that band, and tight against its upper edge**. The build asserts it: lengthen the geometry or edit the band and the two stop being compatible and the build stops. The sight distance and the minutes per setup are chosen by this proposal and registered as `A-SURVEY-001`; the route lengths, the walking speed and the hour band are all already published here.

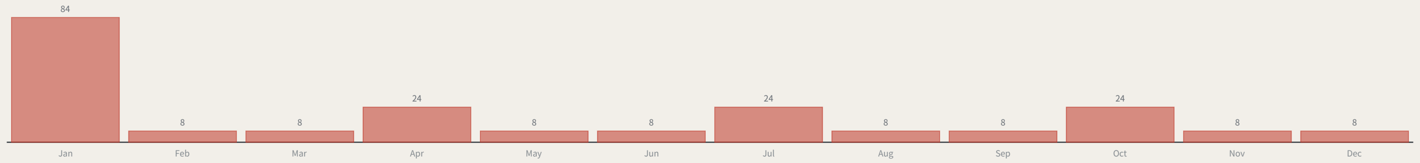
The first order was the last tier without a sheet, and it is not really a place but **an event**: once a year a closing route is run, and this proposal' s measure of trust comes out of that run. From the shipped route lengths, a 50 m sight and 5 minutes a setup, the two routes need 52–55 setups and 5.5–5.8 h — **inside the 4.0–6.0 h the cost table sets by hand** — and the build asserts the two stay compatible.

FIG.30

a type drawing; it carries no orientation

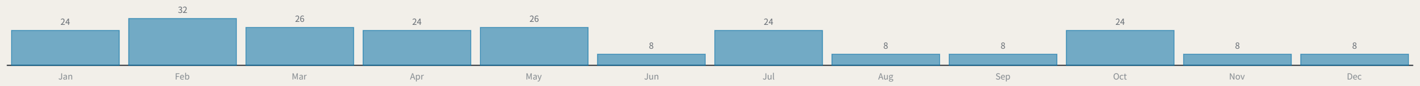
I. The obvious arrangement: every cadence starting in the same month

peak 83.5 h • trough 7.5 h



II. Separated: the same sessions, the same hours, the same table

peak 31.5 h • trough 7.5 h



III. This is a calendar problem, not a resourcing one

Three third-order points monthly, two second-order quarterly, three annual points once each — 8 points, 47 readings. Start every cadence in the same month and the year is not flat: **83.5 h falls in one month against 7.5 h in each of eight others**, eleven to one. The annual table says 139–220 h, which is exactly right and completely unable to say this. A team sized for the average cannot do that month; a team sized for it is idle for eleven. Separating the three annual readings into three months drops the peak to **31.5 h** — and **the search corrected the guess that produced this sheet**: intuition said stagger the quarterly points, and the search leaves them where they are and separates the annual ones instead. The reason is the per-session hours: a datum session is 24 h and a third-order one 2.5 h, so the peak is made almost entirely by annual points landing together, and moving quarterly points around a peak that size changes nothing. **The work does not change at all**: the same 47 sessions, the same hours, the same cost table. The arrangement is searched rather than asserted — every offset combination evaluated, the lowest peak taken — and if separating them did not materially beat the obvious arrangement the build would refuse, because a sheet proposing a schedule that does not help is a sheet arguing for itself.

FIG.29 priced **one** closure and the cost table prices **a year** — and no sheet had drawn the year. Drawn, it shows that with every cadence starting in the same month the peak is 83.5 h against a 7.5 h trough, eleven to one: a team sized for the average cannot do the peak month, and one sized for the peak is idle the rest of the year. Separating the three annual readings drops it to 31.5 h with **no change to the work at all**. This is a calendar problem, not a resourcing one.

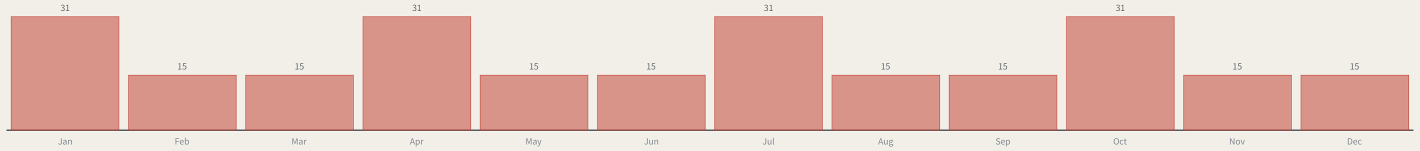
The other year: the hours this network asks of people it does not pay

FIG.31

a type drawing; it carries no orientation

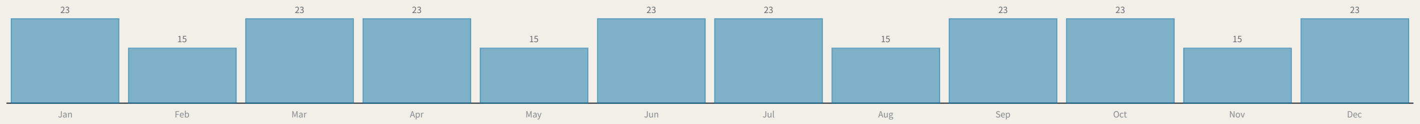
I. Under FIG.30 arrangement: the paid year is flat, the volunteer year is not

peak 31.0 h • trough 15.0 h



II. Separating the quarterly points: the volunteer peak comes down

peak 23.0 h • trough 15.0 h



III. This is not free, and the price is stated here

Volunteers do not attend every session: the datum and first-order runs are professional work with no community places, and community readings fall on the **second and third order** — exactly the tier FIG.30 had no reason to move. So the arrangement that flattens paid hours leaves the volunteer year at 31.0 h against a 15.0 h trough, while that sheet stated that moving the quarterly points changes nothing — **true of the quantity it was measuring and false of this one**. Separating them brings the volunteer peak to 23.0 h at the cost of raising the paid peak from 31.5 h to 33.5 h. **No arrangement minimises both.** Drawing two profiles without saying which is taken hands the decision to the reader while looking rigorous; **this proposal takes the volunteer peak**. A paid peak is a procurement problem — hire, subcontract, move a date. A volunteer peak is a participation failure: community readers cannot be scaled to a month, and that they turn up at all is the floor the whole argument stands on. The 2.0 h added to the paid peak is the price of that choice, stated rather than absorbed.

FIG.30 flattened **paid** hours and said the quarterly points need not move. Volunteers attend only the second and third order — precisely the tier left alone — so the volunteer year stayed at 31.0 h against 15.0 h. Separating them brings it to 23.0 h at a cost of 2.0 h on the paid peak. **Both cannot be minimised at once, and this proposal takes the volunteer peak**: a paid peak is a procurement problem, a volunteer peak is a participation failure.

How many people: the cheapest roster is the one that empties the instrument

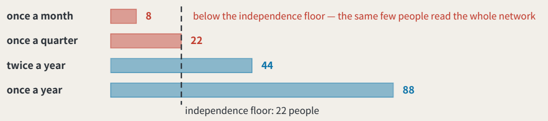
FIG.32

a type drawing; it carries no orientation

I. Eighty-eight attendances, and where they fall

Benchmark	Order	Attendances/yr	Min. distinct readers
BM-21	second	8	2
BM-22	second	8	2
BM-301	third	24	6
BM-302	third	24	6
BM-303	third	24	6
Total		88	22

II. Roster size depends on how often each person is asked



III. The cheapest arrangement is the one that empties the instrument

Community readings fall on the second and third order: **88 attendances** a year, 148–244 h. The cheapest cover is 8 people at one attendance a month — **and that is exactly the failure `A-CLOSURE-002` describes**: this package' s own most fragile assumption states that if the parties are not independent the closure error comes out **systematically small**, and the mechanism reports success having measured nothing. It fails invisibly, and in the direction that looks like success. **So the cheapest staffing answer is the one that empties the instrument.** This sheet turns that into a number: **no community reader is asked for more than 4 attendances a year** — not chosen here but read from `personas.json`, where P3 is the mainstay of the quarterly open days and P8 executes the quarterly re-survey, and nobody is described as coming more often. A third-order point runs 24 attendances and so needs 6 distinct readers; a second-order point runs 8 and needs 2. The network floor is **22 distinct community readers** — not a target but a **floor**, below which the independence this mechanism rests on stops being arguable. **An earlier version of this sheet chose that cap as half a point' s sessions**, which put 4 readers on a third-order point at 6 visits each — every two months, **more than this package' s own personas say they give**. Inventing a number when a citable one already exists is the shape this register keeps recording; cited instead, `A-ROSTER-001` now registers a reference rather than a chosen ratio.

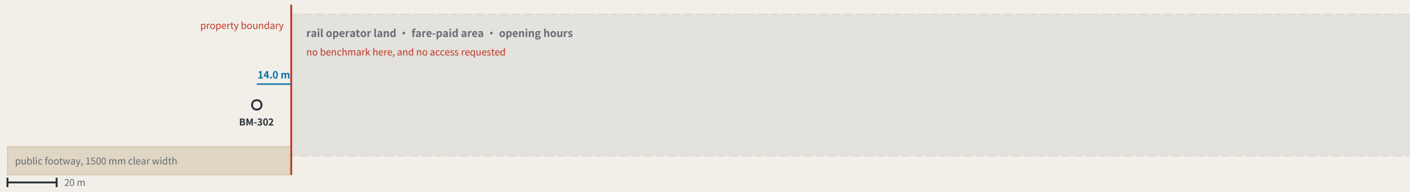
FIG.31 scheduled volunteer **hours** and never asked how many **people** that is. The year needs 88 community attendances, and the cheapest cover is 8 people once a month — **which is exactly the failure A-CLOSURE-002 sets out**: parties that are not independent make the closure error systematically small, and the mechanism reports success having measured nothing. The rule: nobody attends more than half a benchmark' s readings in a year, giving a network floor of 22 people.

Dazhongsi: where the belt meets ground this network does not own

FIG.33

north is up

I. The interface: the point stands outside controlled ground



The 14.0 m is not a rule invented here: the prose already fixes a 14 m setback for buildings along the spine, and the benchmarks sit on the axis, so the nearest point stands clear of controlled ground as a consequence. **The build checks that no benchmark falls inside any building footprint** — one that did would be a reading that needs a ticket.

II. Handover: what crosses the boundary is the reading, not the instrument

Item	Rule	Why
What crosses	**the reading, not the instrument**	the operator levels their own point on their own ground, to the same tolerance class, and hands over a number
What does not	**access, equipment, working hours**	this proposal does not ask for access it cannot be granted, nor pretend a point it cannot reach is a point it holds
If they decline	**the segment is recorded **unread****	not estimated, not interpolated, not assumed within tolerance — an unread segment is a visible hole in the closure record
What the hole means	**that segment may not be used to claim compliance**	**which is precisely why this package built a closure record that can show a hole**
Where a point may stand	**never on controlled ground**	if a reading needs a ticket, "anyone can walk up to a benchmark" is false as stated

Every other sheet assumes this network reaches its own points; at the station that ends — the landowner is not the city. Three rules: **no benchmark on controlled ground** (BM-302 stands 14.0 m clear of the hub, checked at build time); **what crosses is the reading, not the instrument**; and **if the operator declines the segment is recorded unread** — not estimated, not interpolated, a visible hole in the closure record.